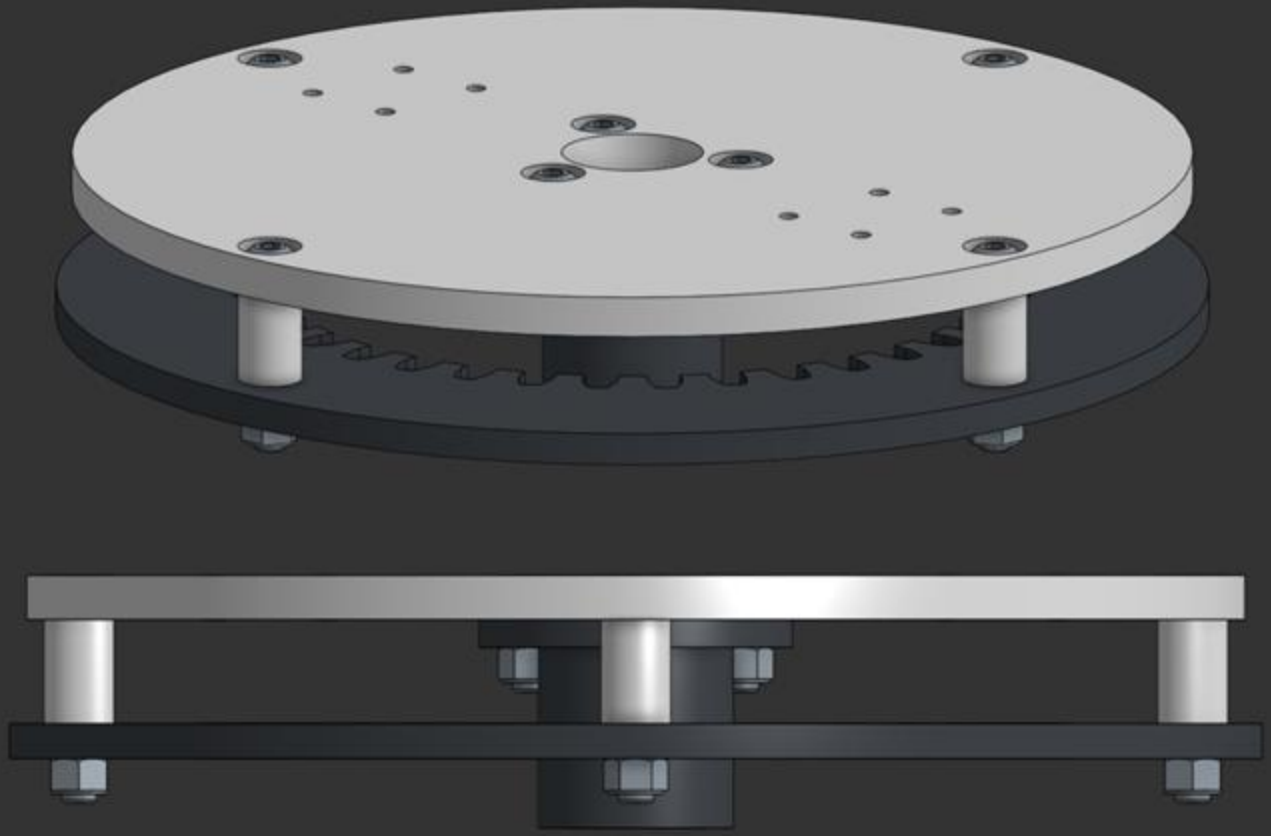
Arm Base Plate Assembly Instructions

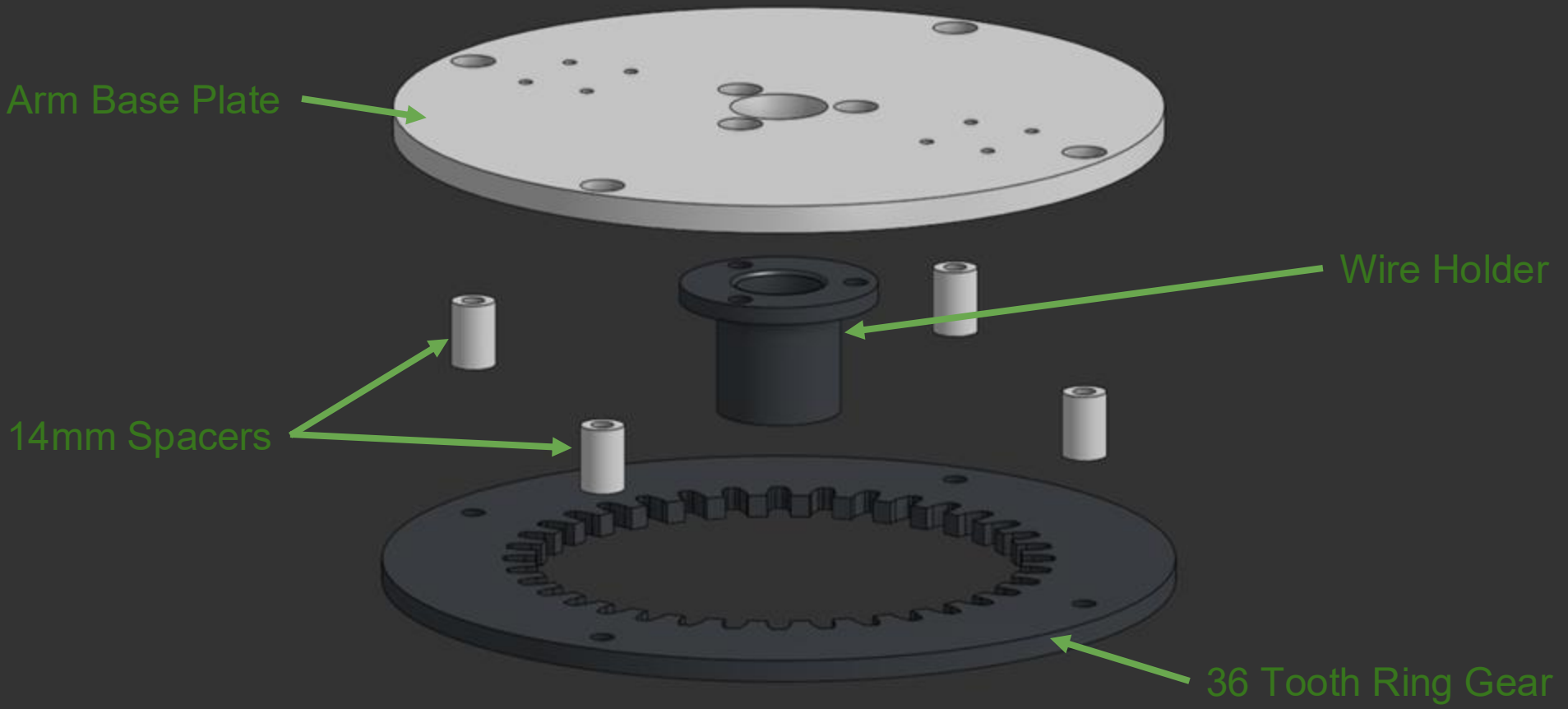


Parts

Major Components:
Arm Base Plate
Wire Holder
36T Ring Gear
4x M5x10x20mm Spacer

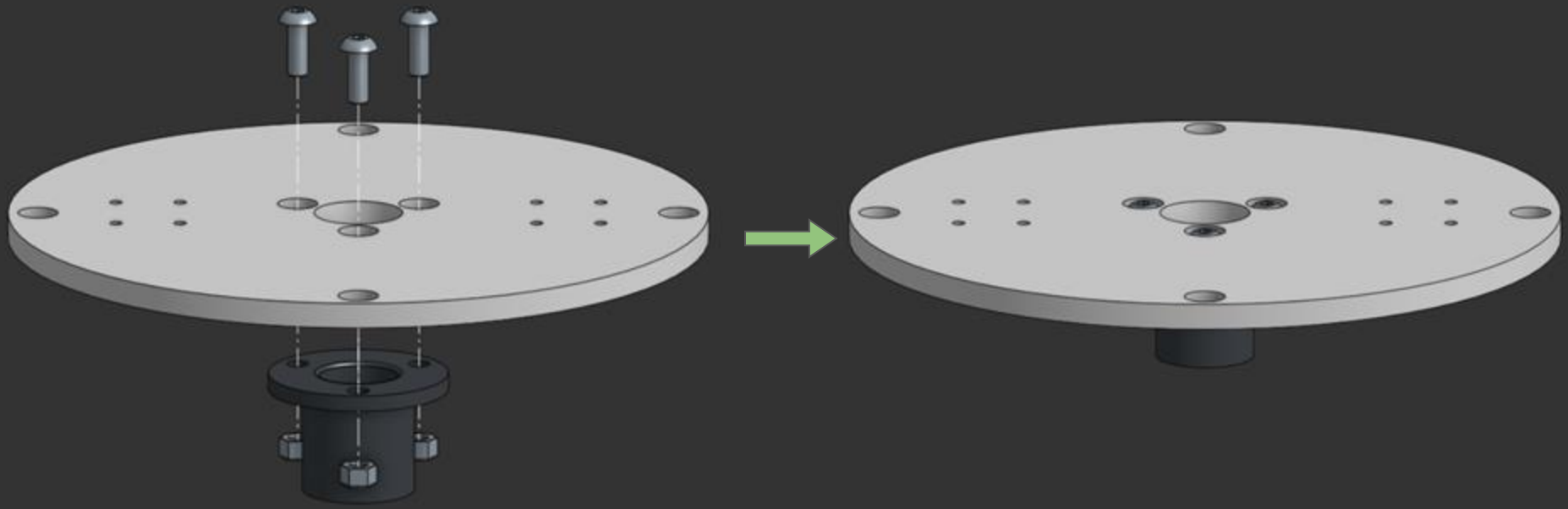
Minor Components:
4x M5x30mm Button Head Bolt
3x M5x14mm Button Head Bolt
7x M5 Nylock Nuts

Major Components



Step 1

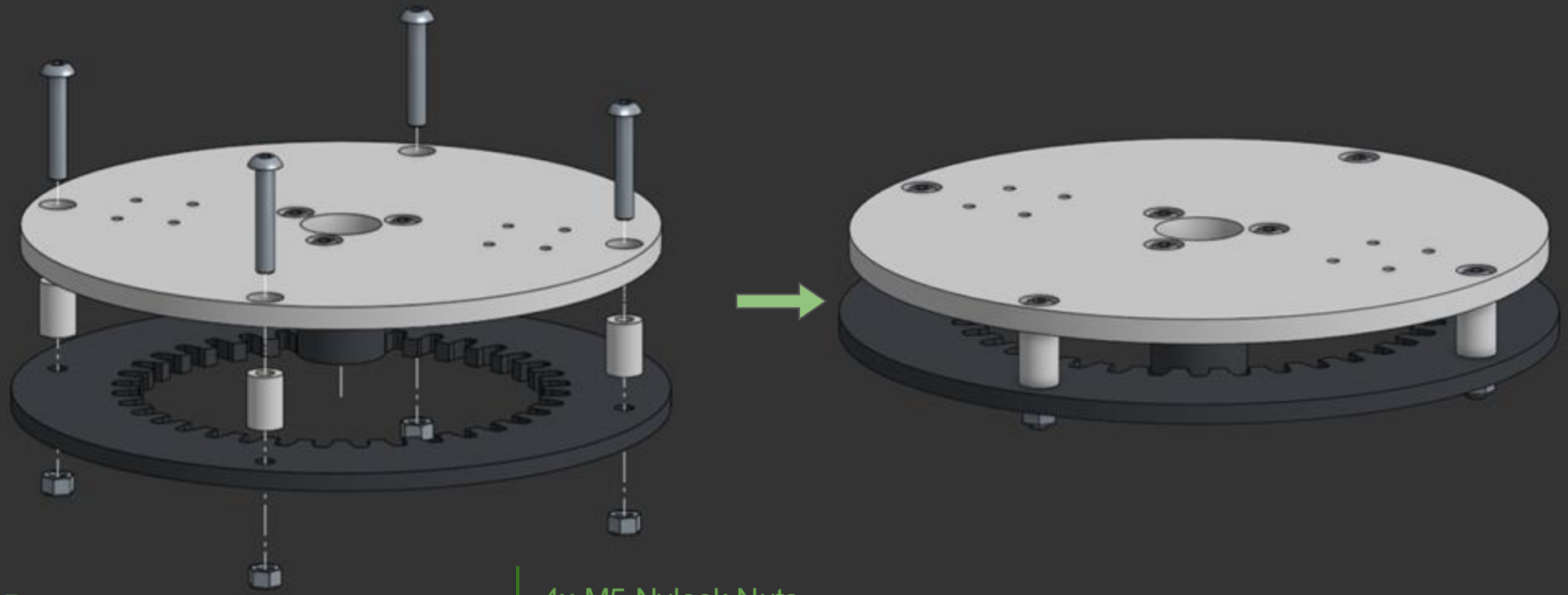
Bolt the Wire Holder through the center of the Arm Base Plate



Parts
-Arm Base Plate
-Wire Holder
-3x M5x14mm Button Head Bolts
-3x M5 Nylock Nuts

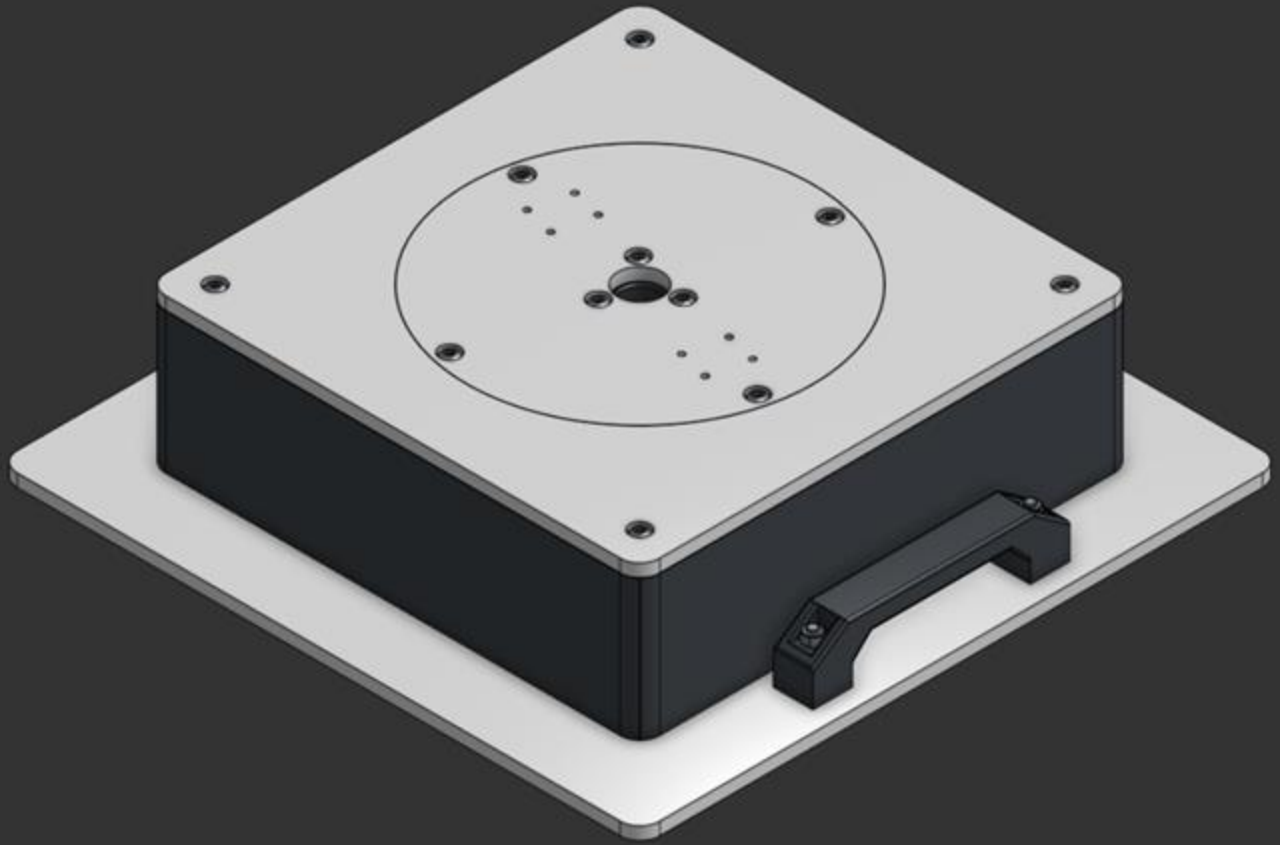
Step 2

Bolt the Arm Base Plate to the 36T Ring Gear with 20mm spacers in between



Parts
-Previous step
-4x M5x10x20mm Spacer
-4x M5x30mm Button Head Bolts
-4x M5 Nylock Nuts
-36T Ring Gear

Arm Base Assembly Instructions



Parts

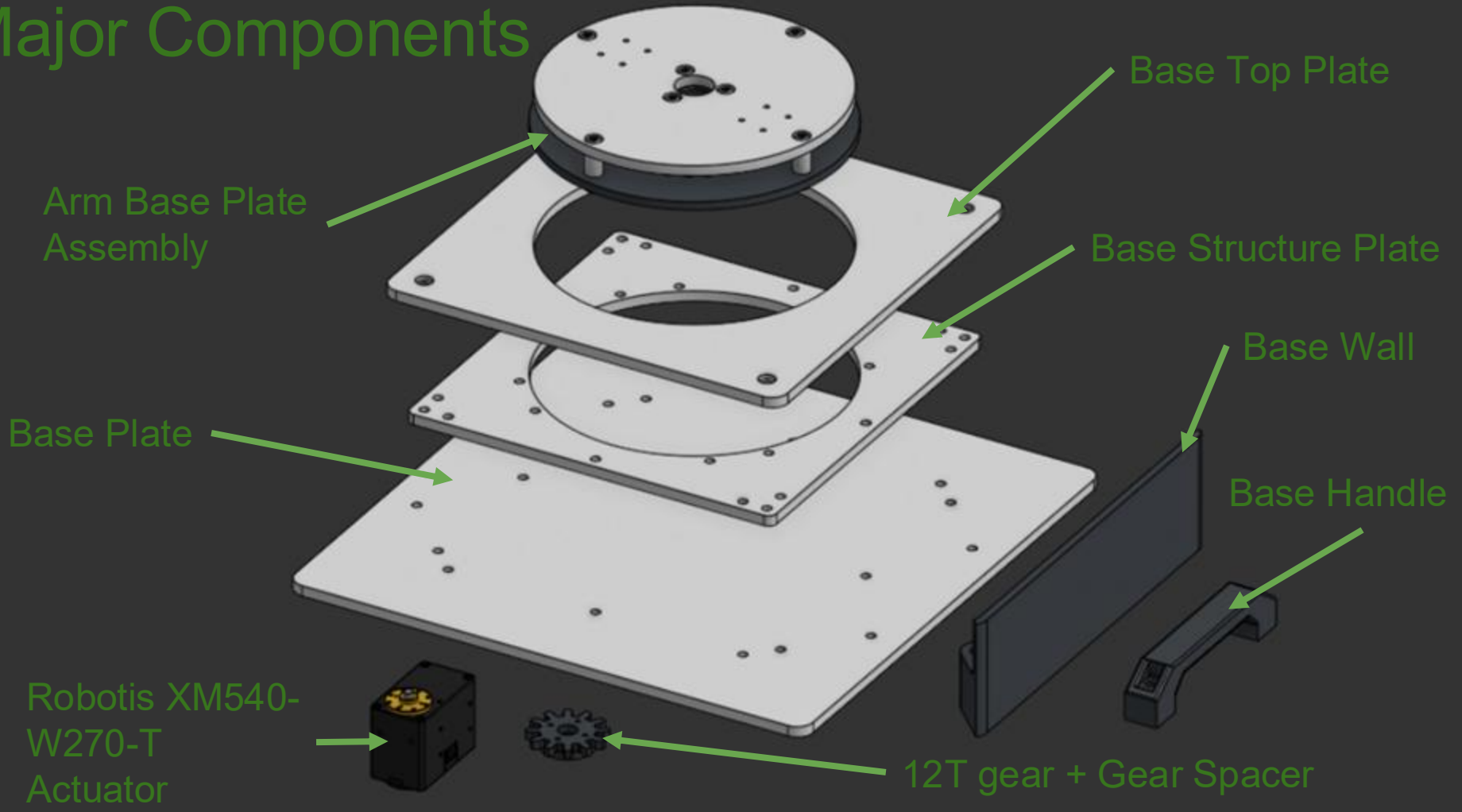
Major Components:
Arm Base Plate Assembly
Base Plate
Structure Plate
Base Top Plate
4x Base Wall
2x Base Handle
12T Ring Gear
Robotis XM540-W270-T
Actuator

Parts Continued

Minor Components:

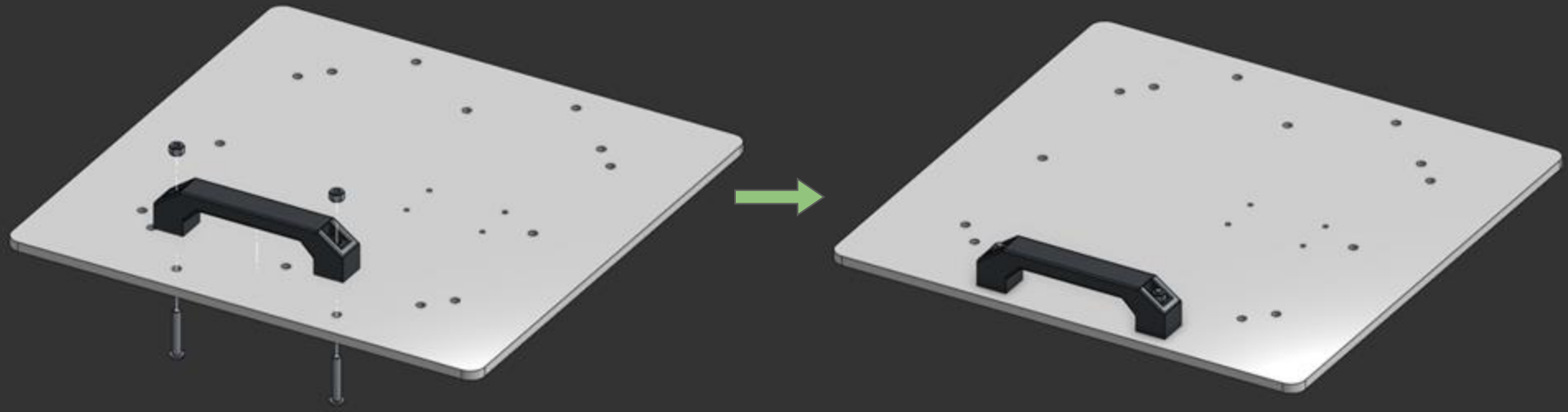
20x Nylock Nut
8x M5x14mm Button Head Bolt
4x M5x30mm Button Head Bolt
8x M5x30mm Socket Head Bolt
8x M5x60mm Button Head Bolt
4x M2.5x12mm Socket Head Bolt
4x M2.5x25mm Socket Head Bolt
8x 5x11x5mm Bearing
16x 5x16x5mm Bearing
4x 5x8x25mm M5 Threaded Spacers
8x 5x10x40mm M5 Threaded Spacers
4x Robotis Spacers

Major Components



Step 1

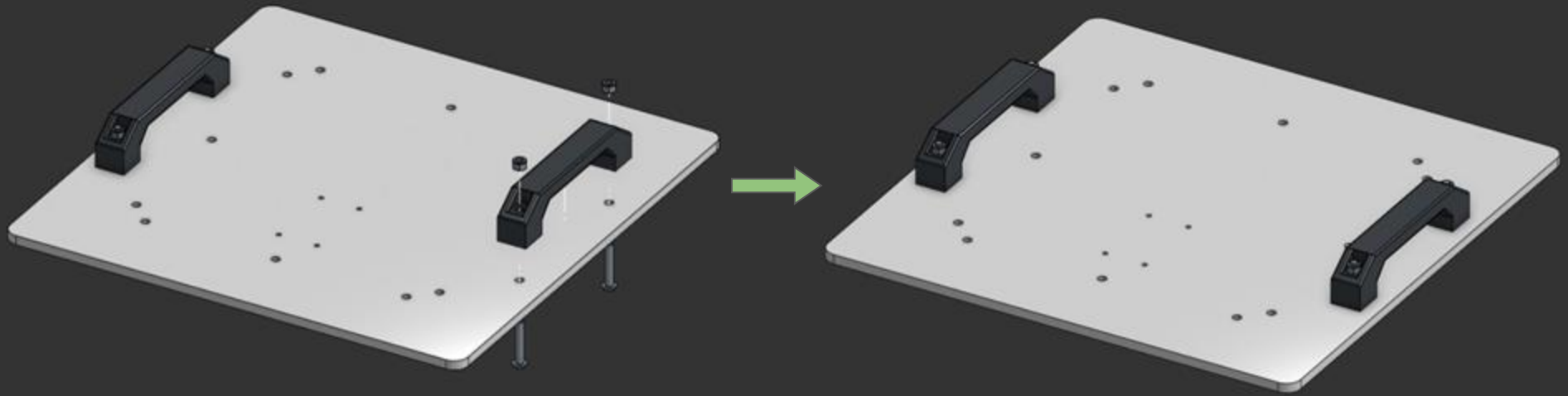
Bolt the Base Handle to the edge of the Base Plate



Parts
-Base Plate
-Base Handle
-2x M5x30mm Button Head Bolt
-2x Nylock Nuts

Step 2

Repeat for the other side

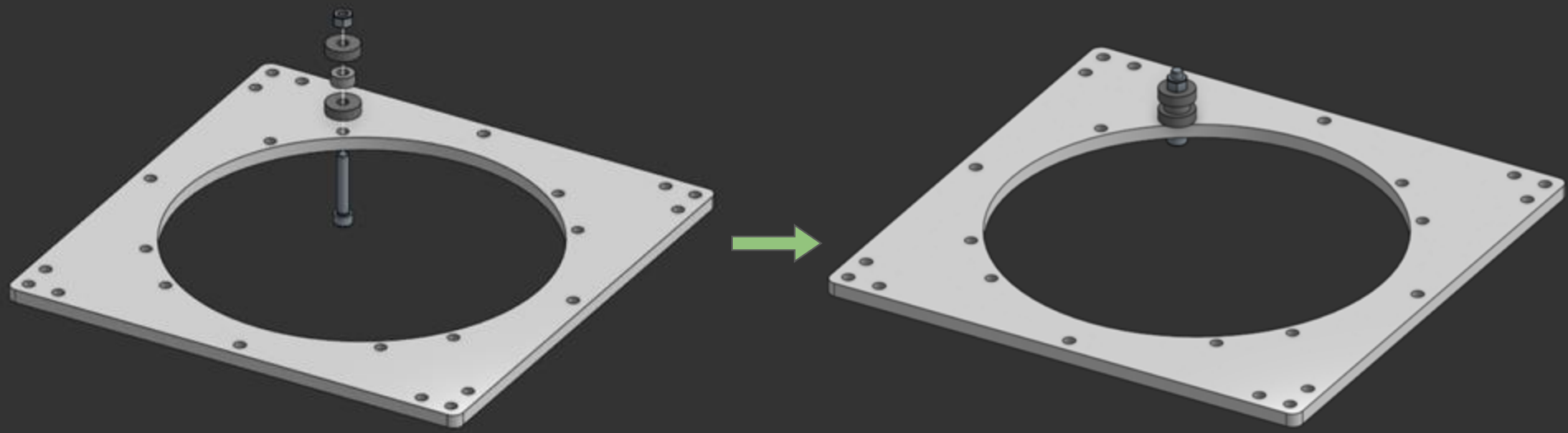


Parts
-Previous step
-Base Handle
-2x M5x30mm Button Head Bolt

-2x Nylock Nuts

Step 3

Bolt a bearing stack (in the order of wide, thin, wide) around the center of the structure plate

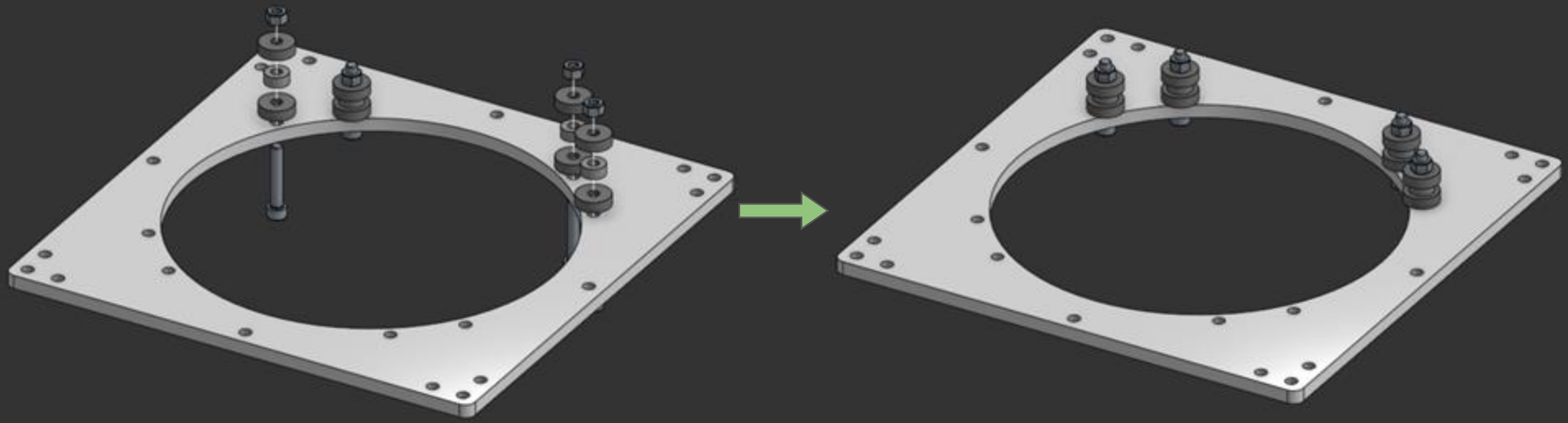


Parts
-Structure Plate
-2x 5x16x5mm Bearing
-5x11x5mm Bearing

-M5x30mm Hex Socket Head Bolt
-Nylock Nut

Step 4

Bolt 3 more bearing stacks around one half of the center of the structure plate

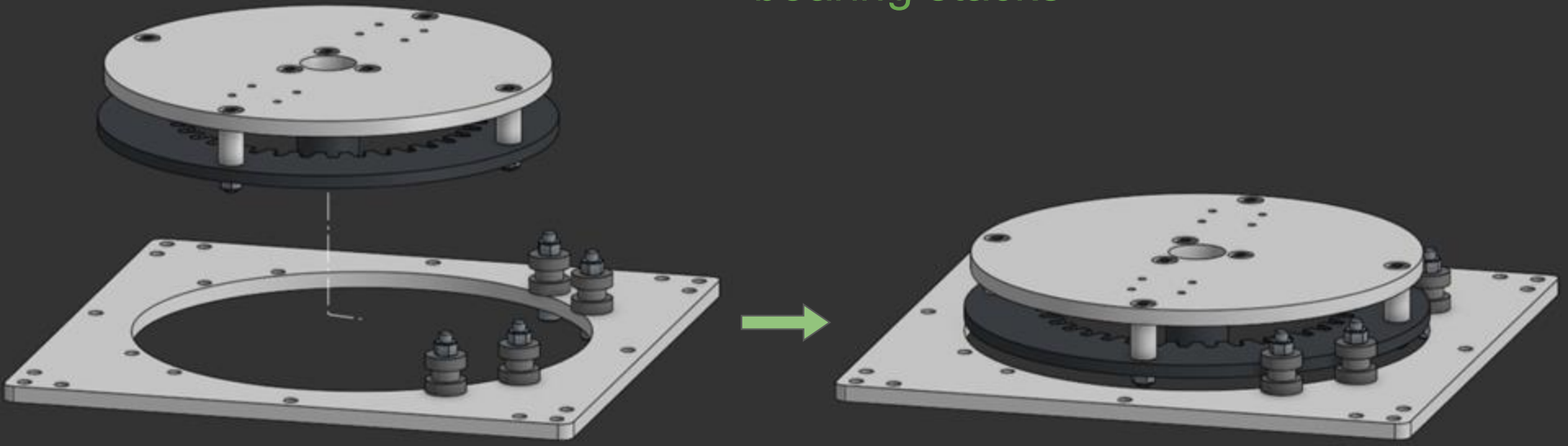


Parts
-Previous step
-6x 5x16x5mm Bearing
-3x 5x11x5mm Bearing

-3x M5x30mm Hex Socket Head Bolt
-3x Nylock Nut

Step 5

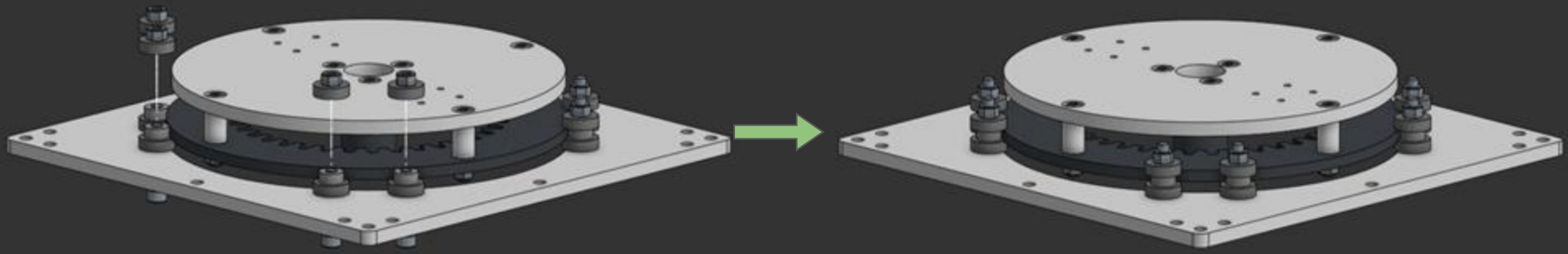
Insert the Arm Base Assembly between the bearings in the bearing stacks



Parts
-Previous step
-Arm Base Assembly

Step 6

Bolt four more bearing stacks around the other half of the structure plate

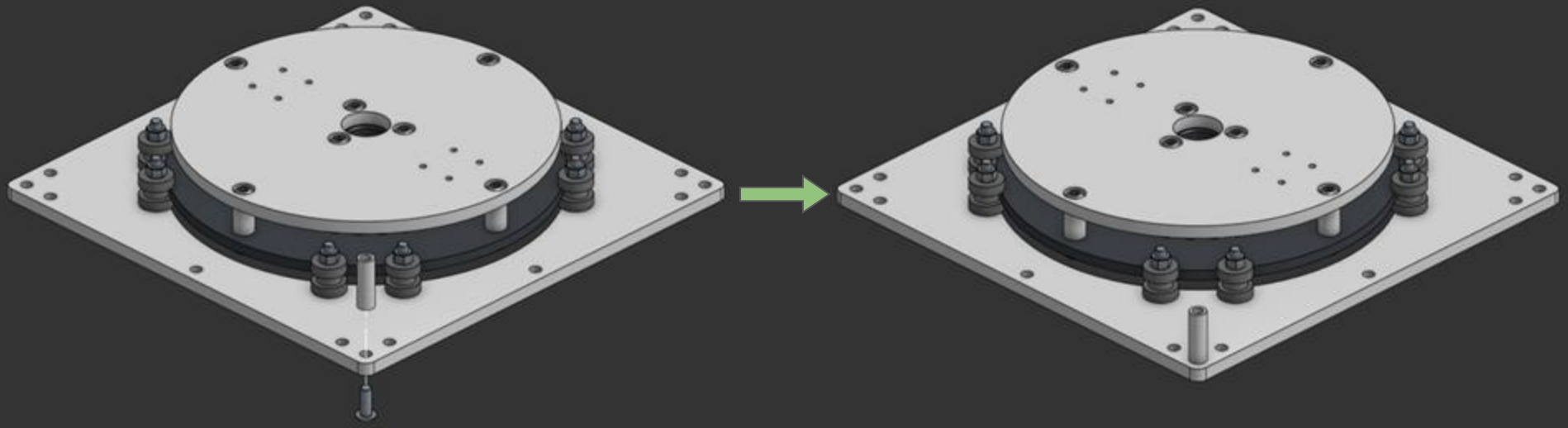


Parts
-Previous step
-8x 5x16x5mm Bearing
-4x 5x11x5mm Bearing

-4x M5x30mm Hex Socket Head Bolt
-4x Nylock Nut

Step 7

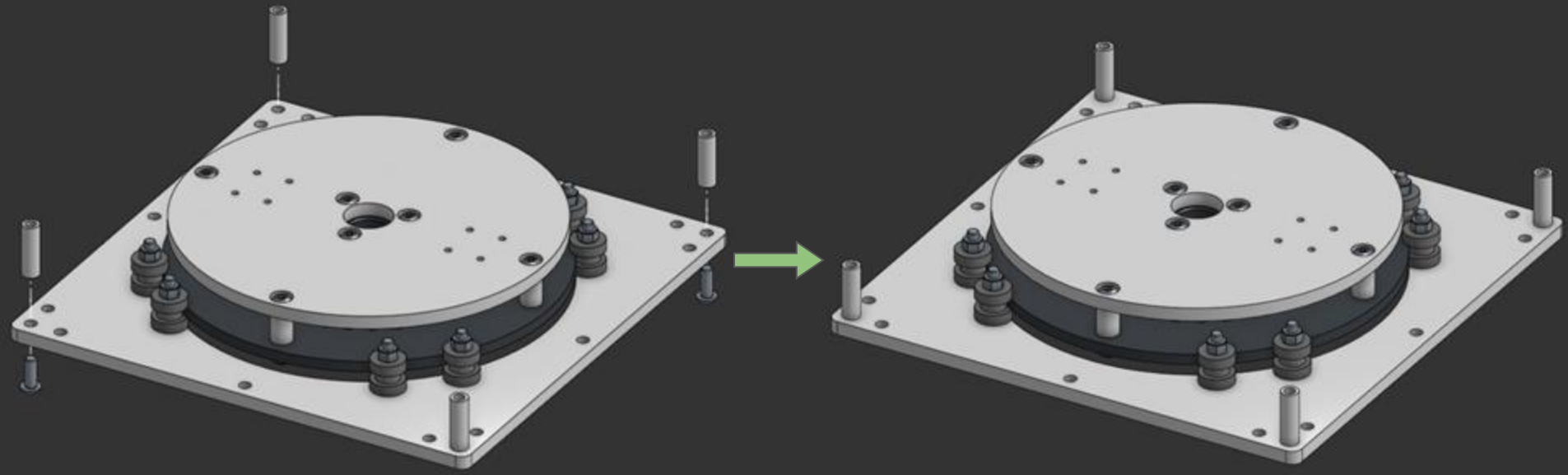
Bolt a threaded spacer onto one corner of the Structure Plate



Parts
-Previous step
-5x8x25mm M5 Threaded Spacers
-M5x14mm Button Head Bolt

Step 8

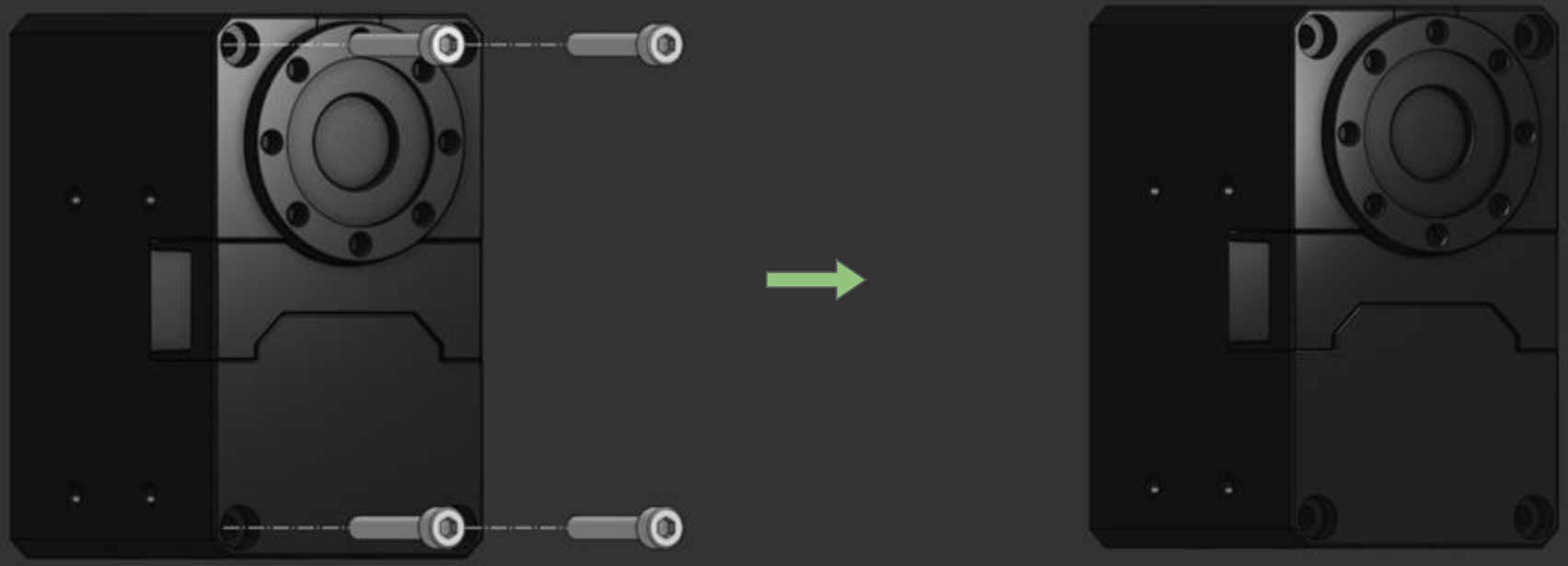
Bolt threaded spacers on all other corners of the Structure Plate



Parts
-Previous step
-3x 5x8x25mm M5 Threaded Spacers
-3x M5x14mm Button Head Bolt

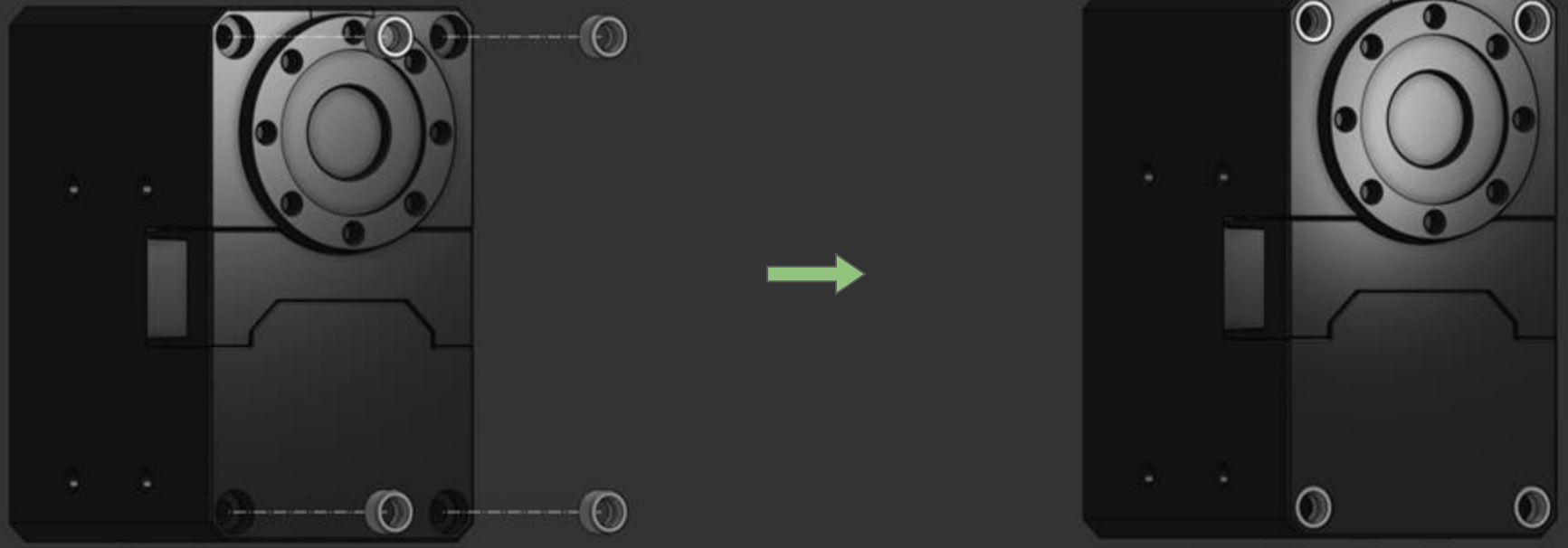
Step 9

Remove the four bolts on the back of the actuator



Parts
-Robotis XM540-W270-T Actuator

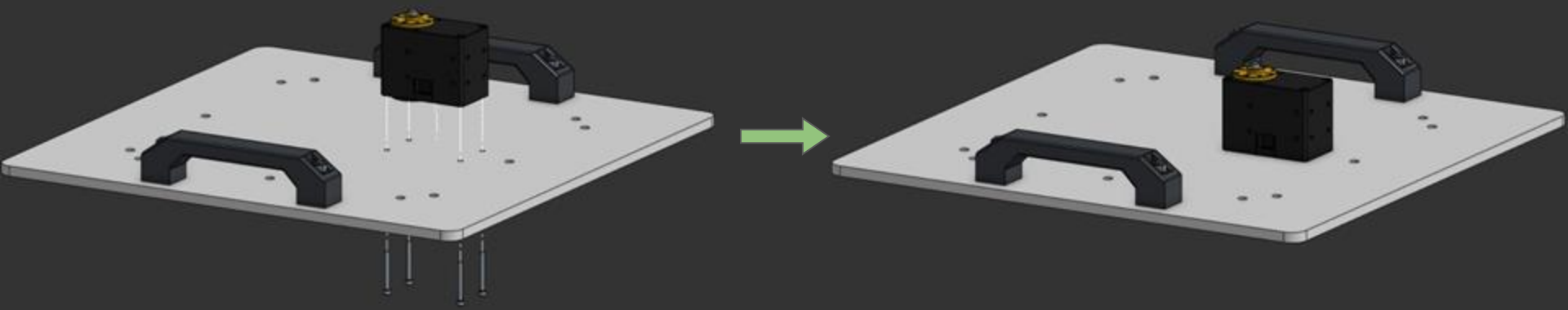
Step 10 Insert four Robotis Spacers into the back of the actuator



Parts
-Robotis XM540-W270-T Actuator
-4x Robotis Spacers

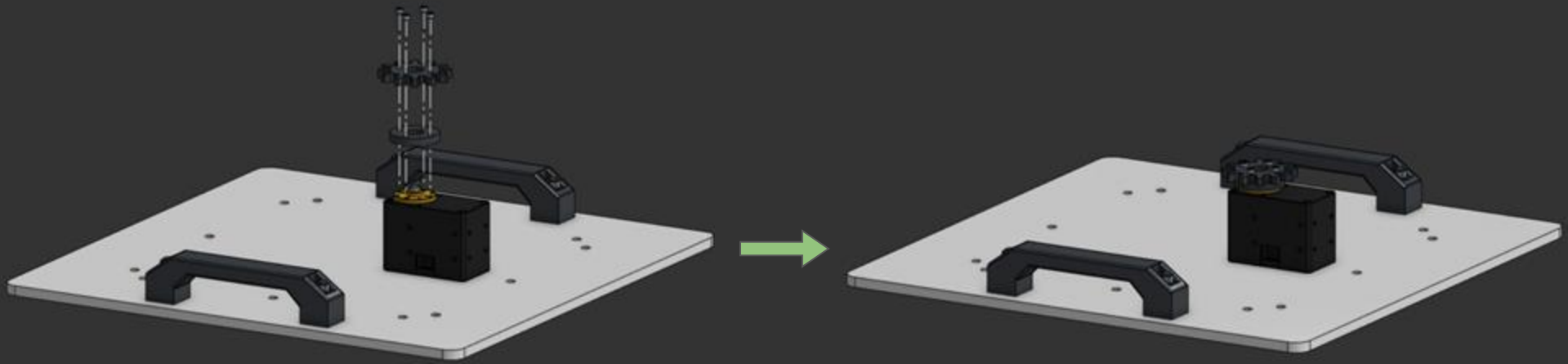
Step 11

Bolt the back of the actuator to the Base Plate



Parts
-Previous step
-Base Plate (with handles)
-4x M2.5x25mm Socket Head Bolt

Step 12 Bolt the 12T Gear on top of the Gear Spacer to the actuator

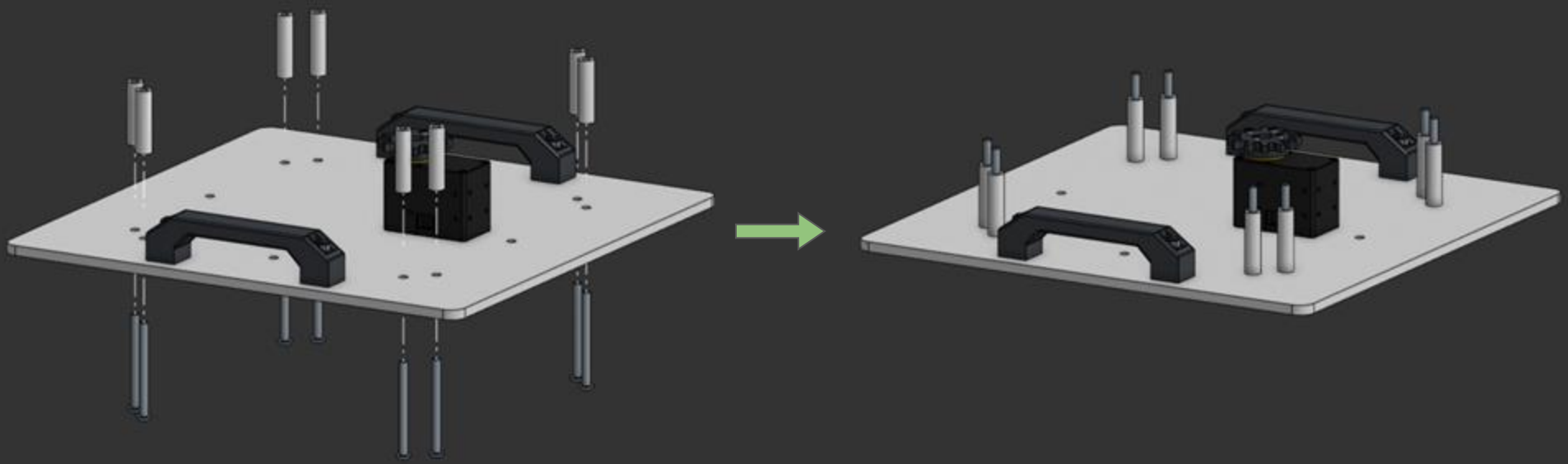


Parts
-Previous step
-12T Gear
-Gear Spacer

-4x M2.5x12mm Socket Head Bolt

Step 13

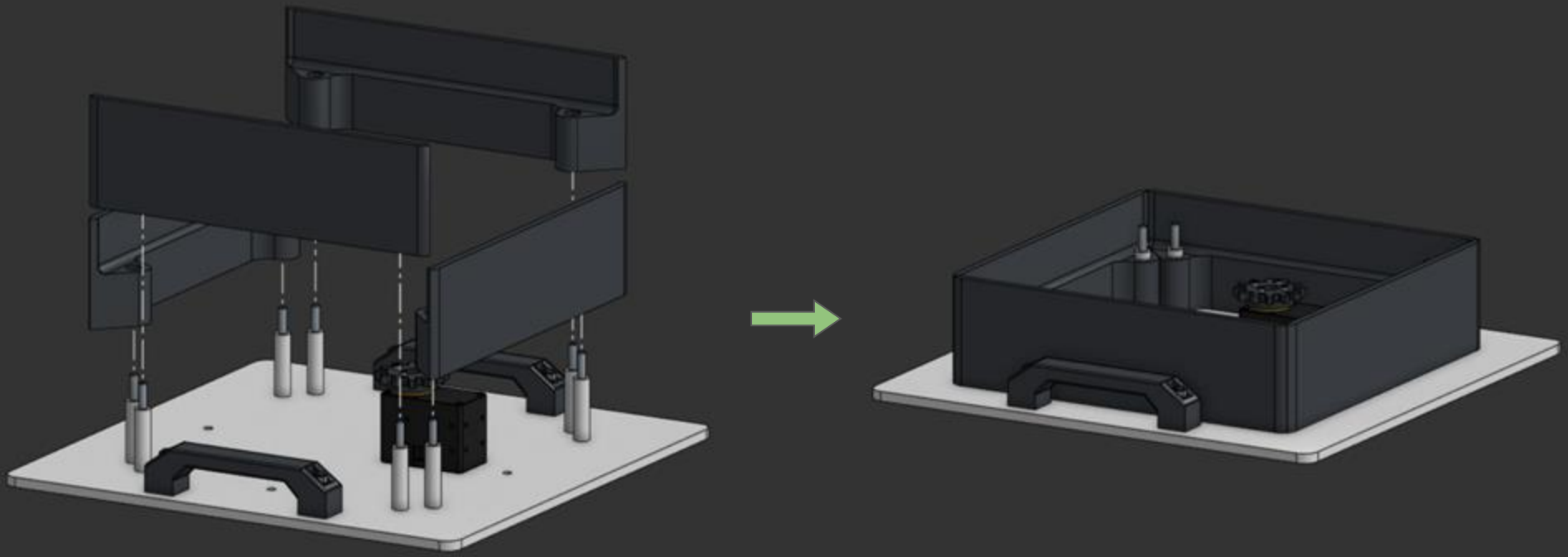
Insert M5x60mm Button Head Bolts through eight corner holes, and place 40mm spacers on top



Parts
-Previous step
-8x M5*60mm Button Head Bolt
-8x 5x10x40mm M5 Threaded Spacers

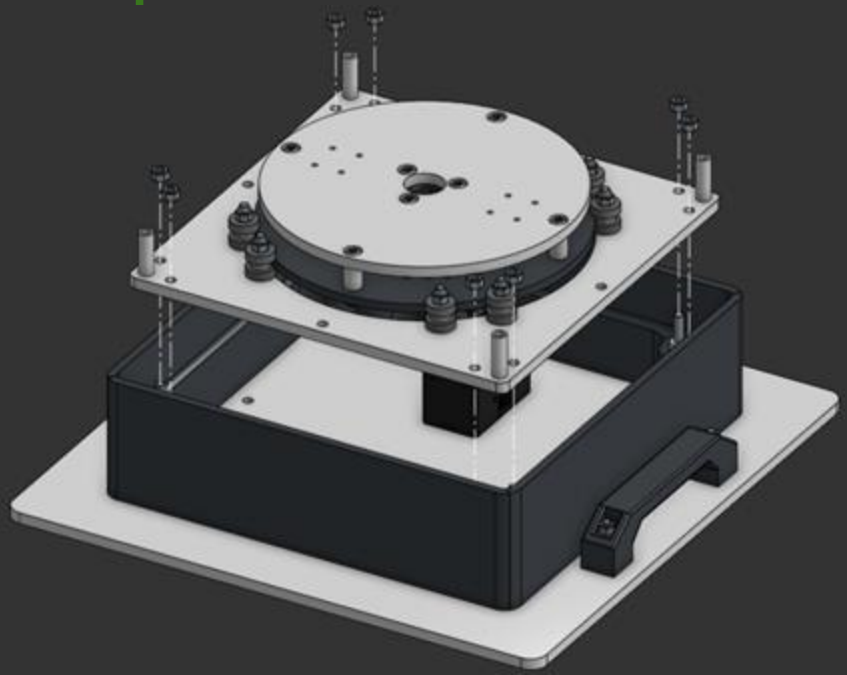
Step 14

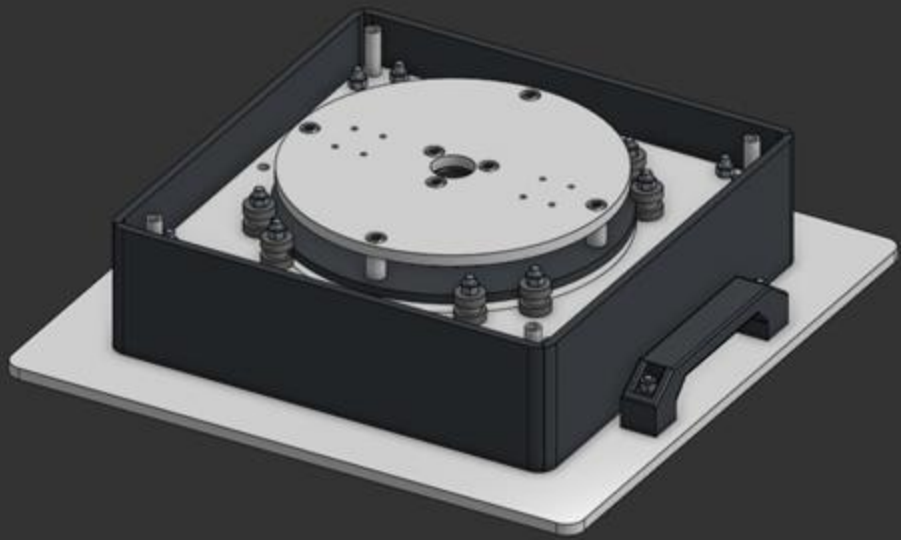
Insert Base Walls onto each face of the base



Parts
-Previous step
-4x Base Wall

Step 15 Insert the Arm Base Plate Assembly on top of the bolts and attach Nylock nuts



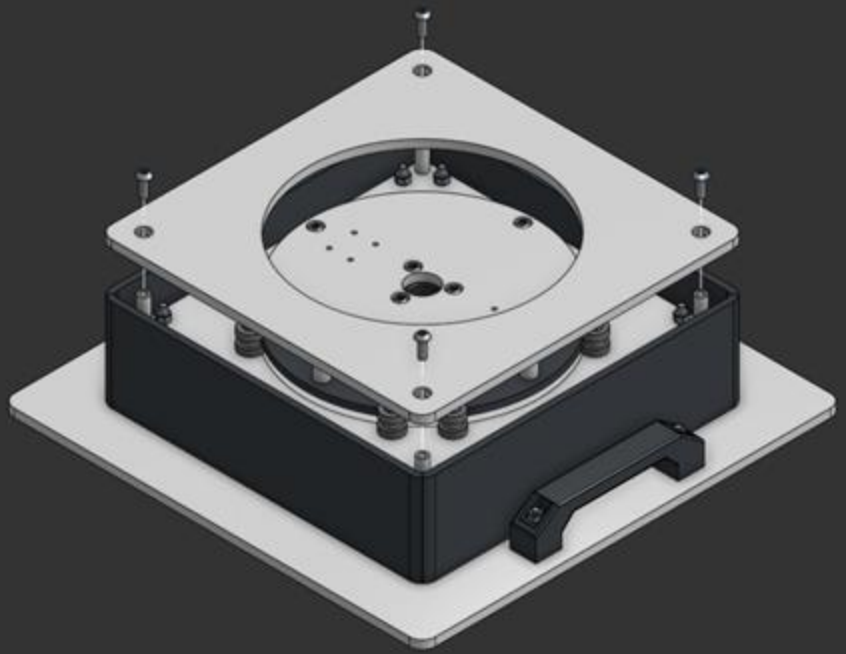


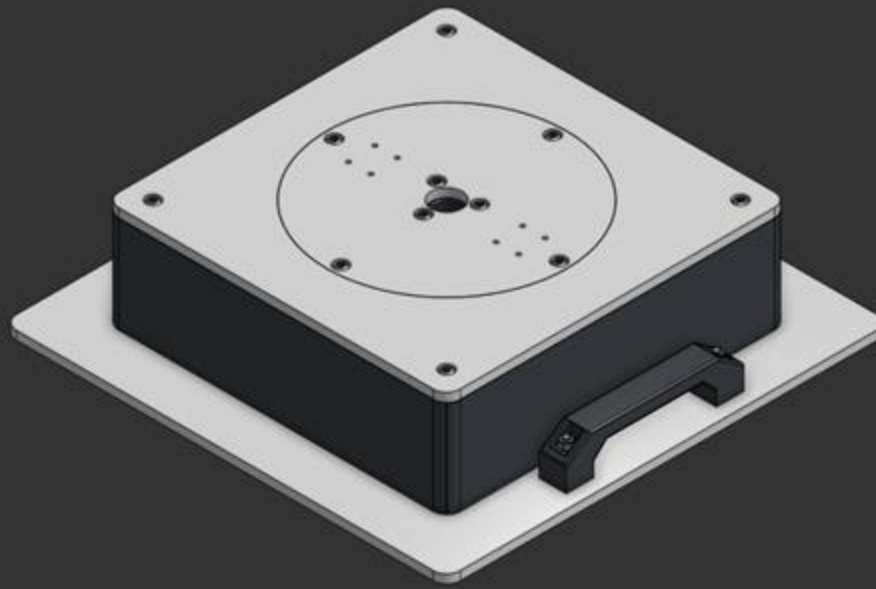
Parts
-Previous step
-Arm Structure Plate and Arm Base Plate Assembly

-8x Nylock Nuts

Step 16

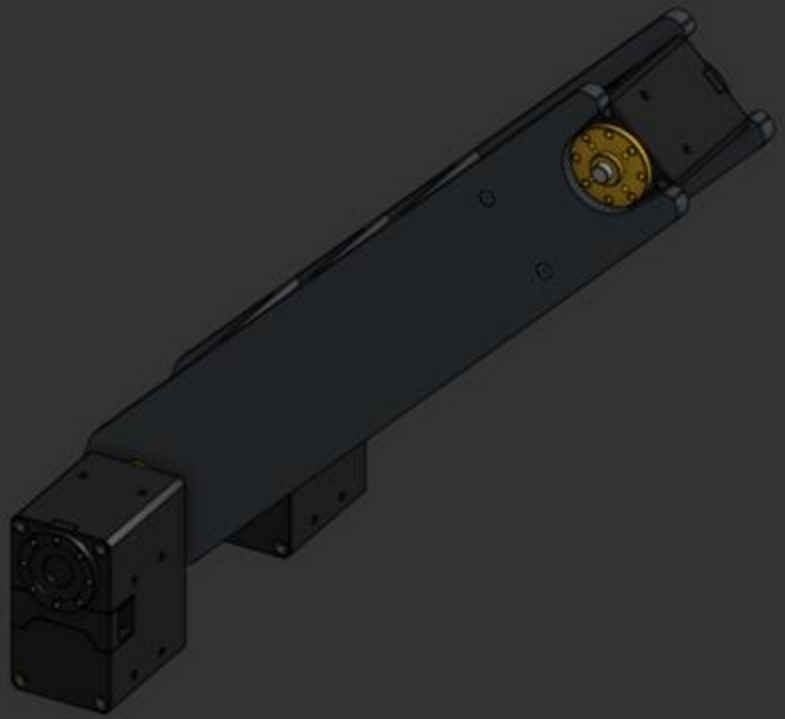
Place the Base Top Plate around the Arm Base Plate and bolt each corner

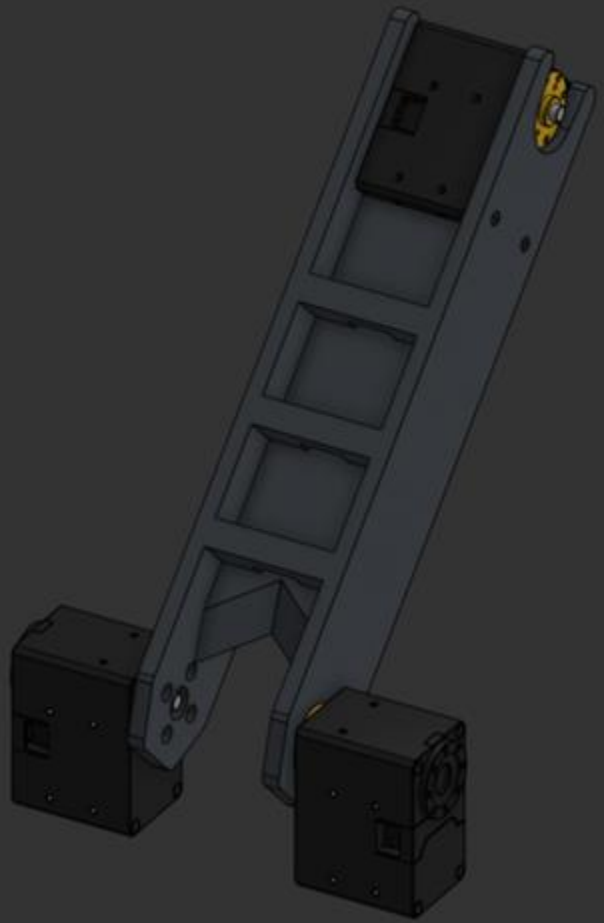




Parts
-Previous step
-4x M5x14mm Button Head Bolt

Arm Shoulder Assembly Instructions



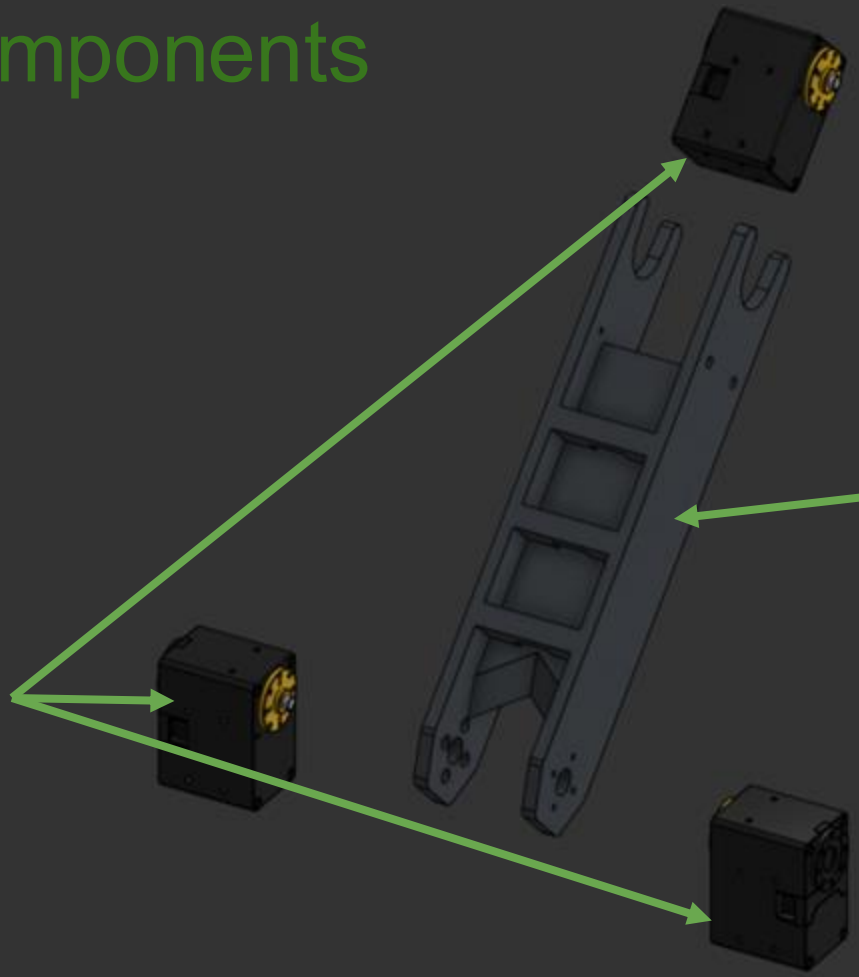


Parts

Major Components:
Arm Shoulder Piece
3x Robotis XM540-W270-T Actuator

Minor Components:
8x M2.5x6mm Bolts
4x M2.5x25mm Bolts
4x Robotis Spacers

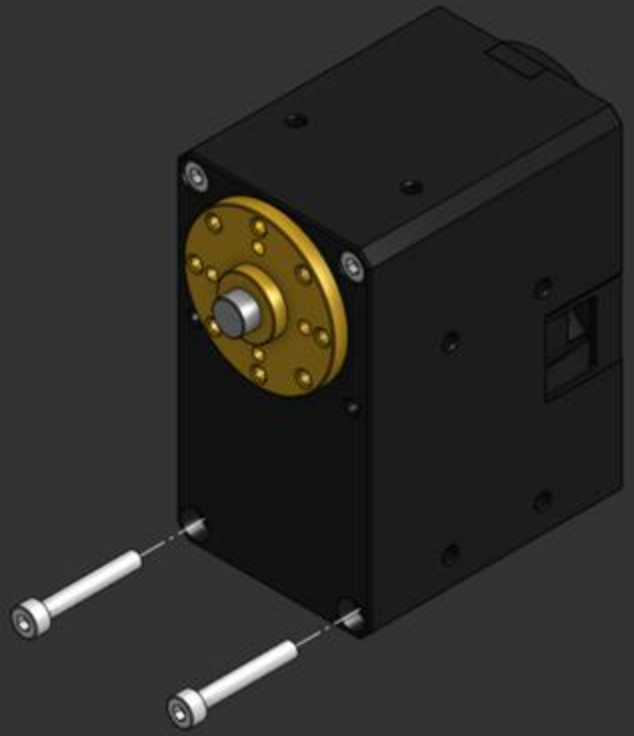
Major Components

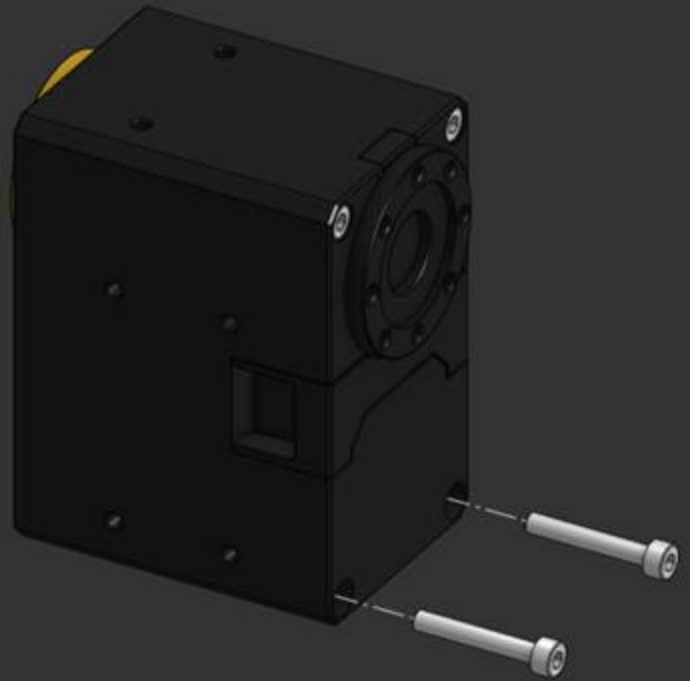


Arm Shoulder Piece

Robotis XM540-W270-T Motors

Step 1 Remove the bottom two bolts from both sides of the actuator

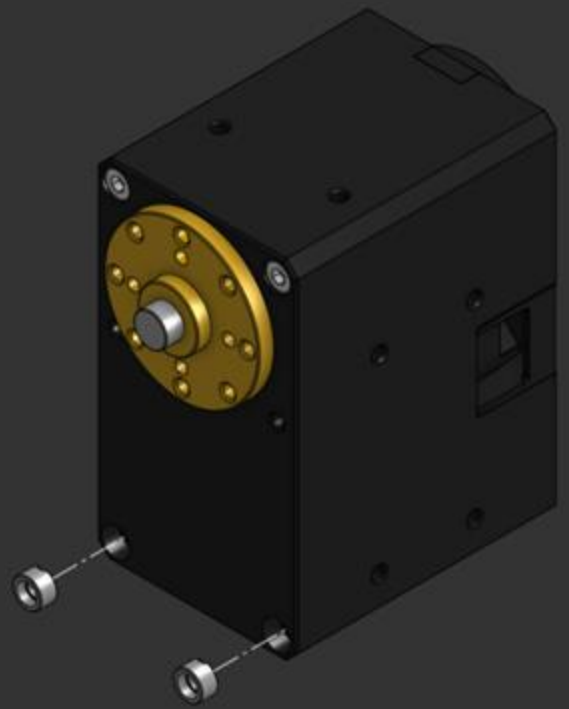




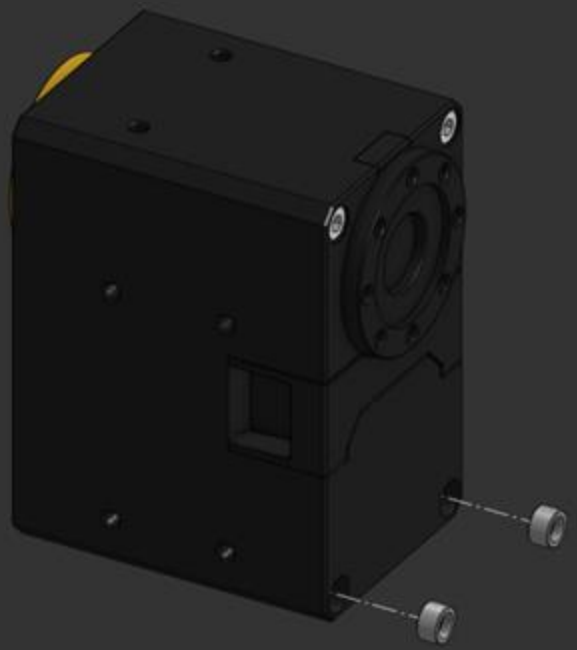
Parts
-Robotis XM540-W270-T Actuator

Step 2

Insert spacer rings into the actuator



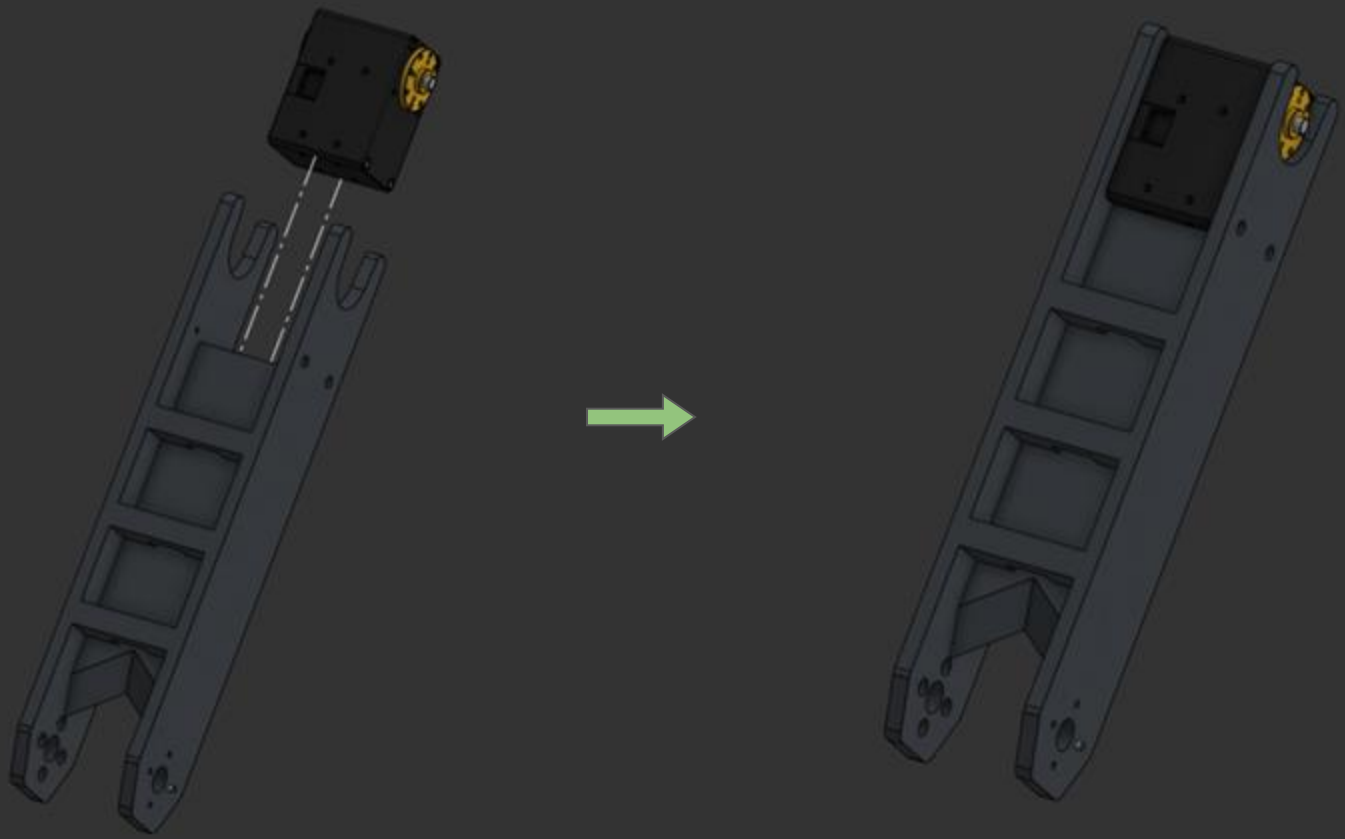




Parts
-Robotis XM540-W270-T Actuator
-4x Robotis Spacers

Step 3

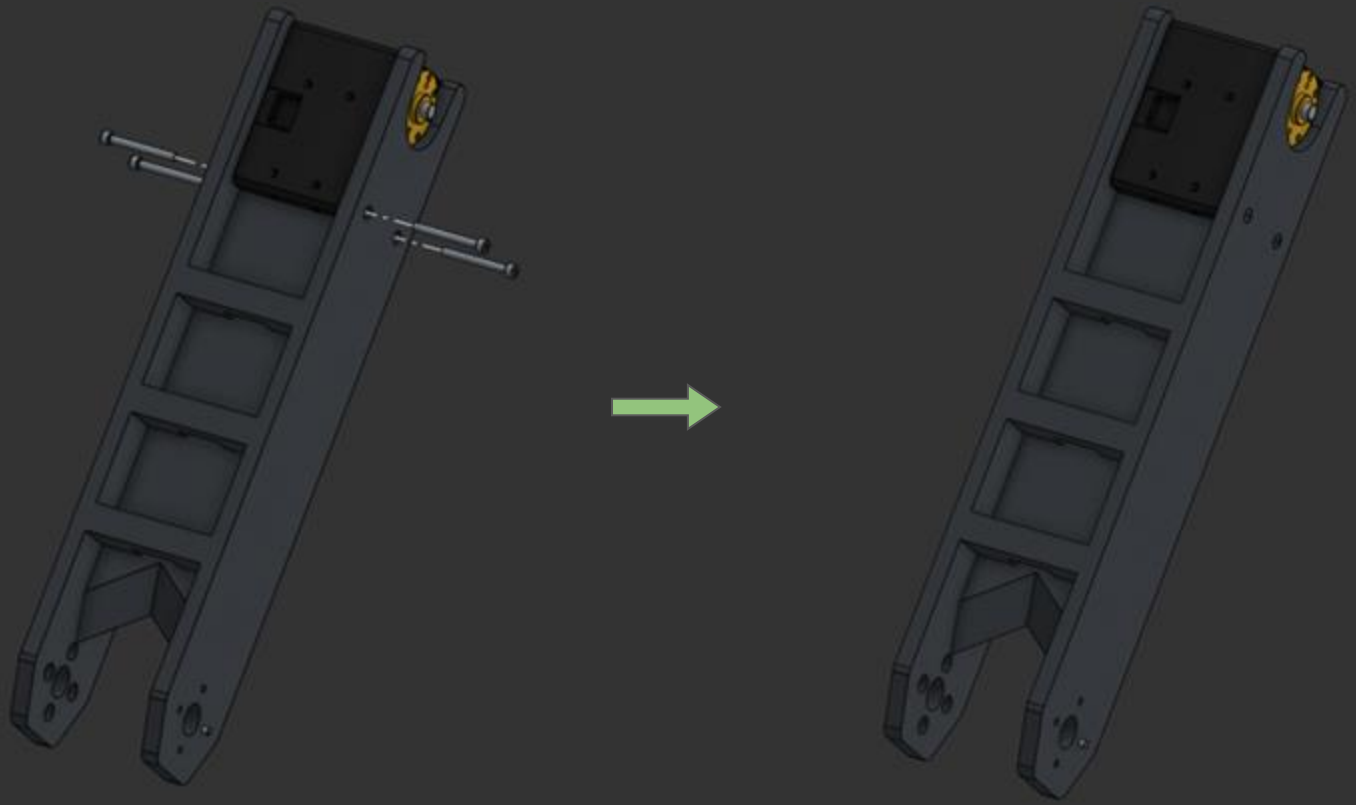
Place the actuator in the top of the shoulder piece



Parts
-Previous step
-Arm Shoulder Piece

Step 4

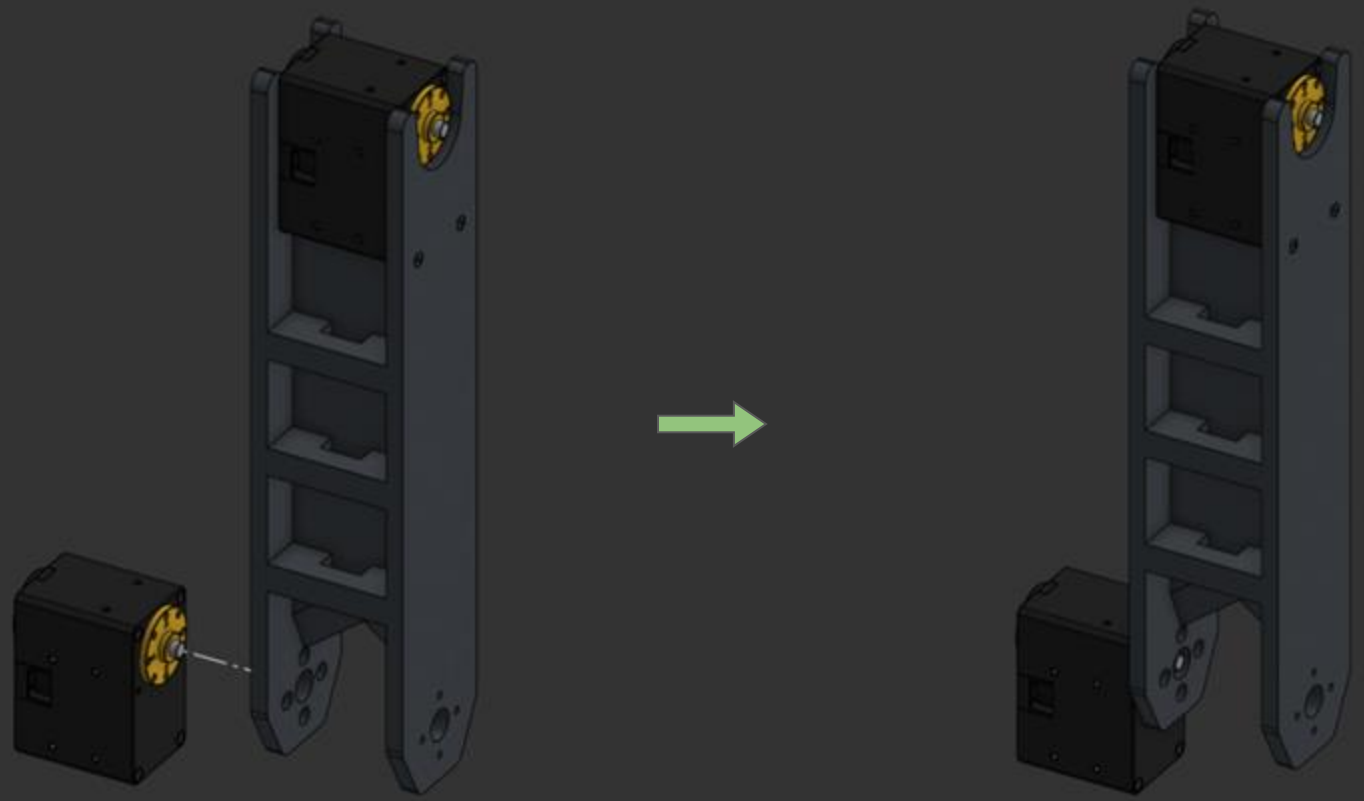
Bolt the actuator into the top of the shoulder piece



Parts
-Previous step
-4x M2.5x25mm Socket Head Bolt

Step 5

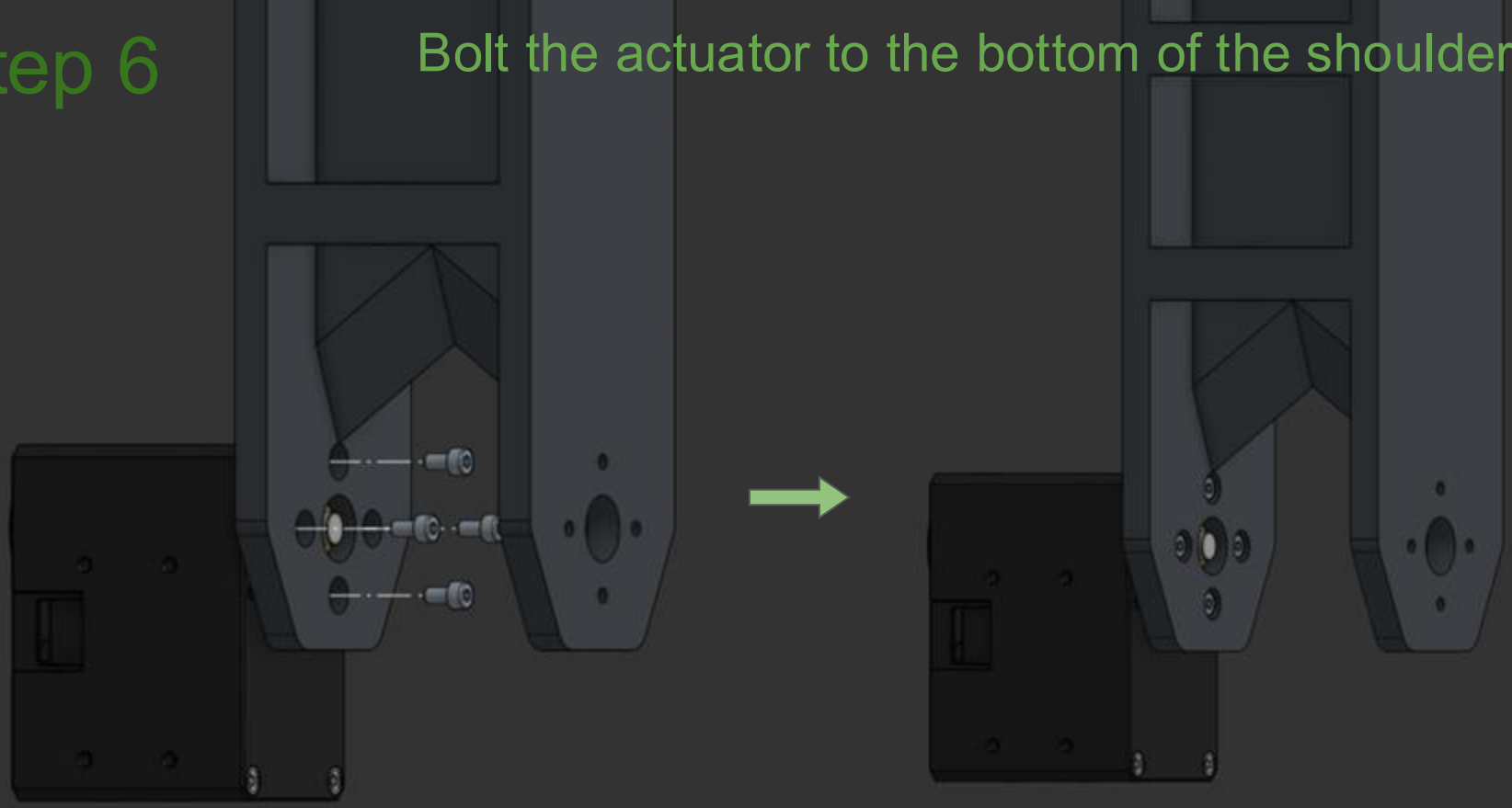
Align the actuator with the bottom of the shoulder piece



Parts
-Previous step
-Robotis XM540-W270-T Actuator

Step 6

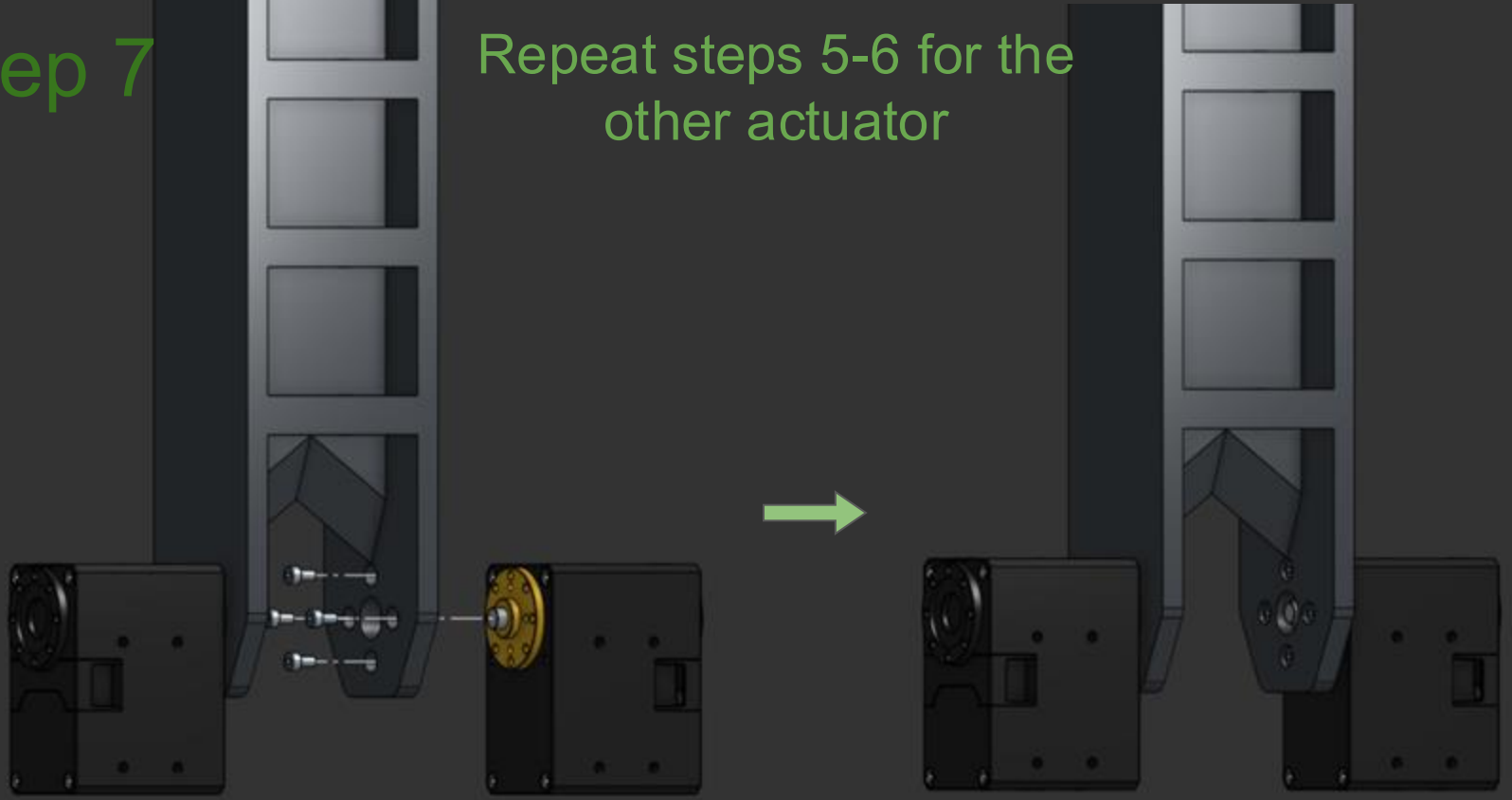
Bolt the actuator to the bottom of the shoulder piece



Parts
-Previous step
-4x M2.5x6mm Bolts

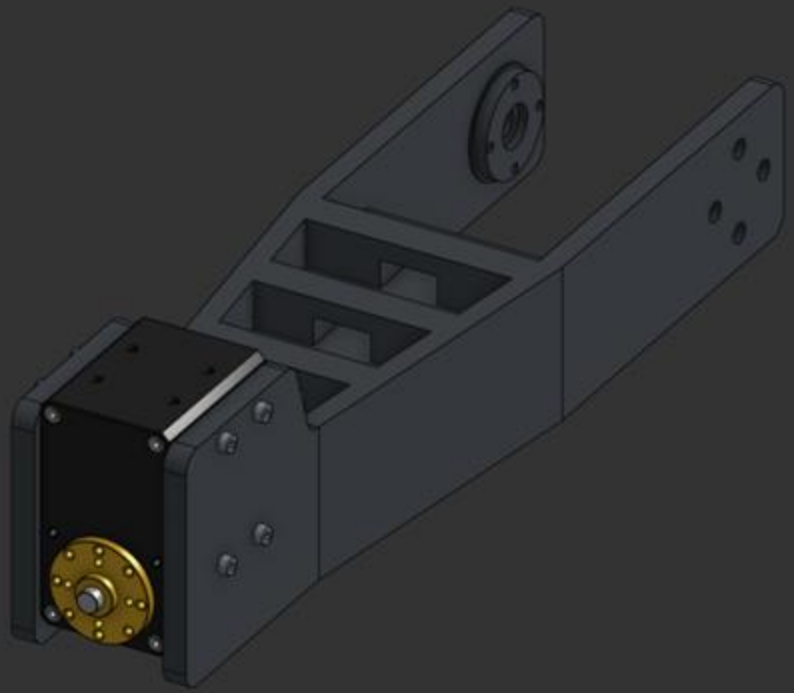
Step 7

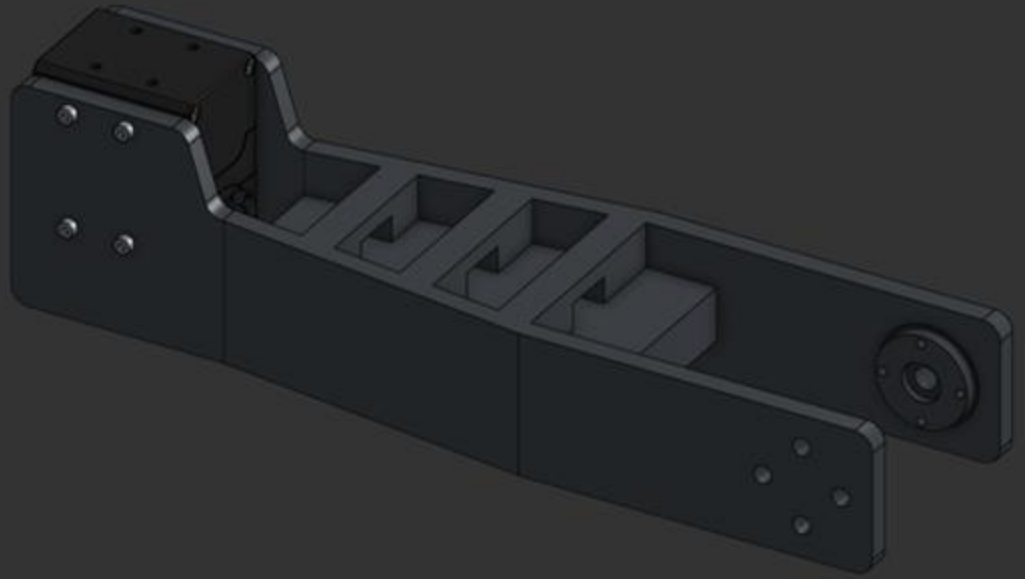
Repeat steps 5-6 for the other actuator



Parts
-Previous step
-Robotis XM540-W270-T Actuator
-4x M2.5x6mm Bolts

Arm Elbow Assembly Instructions





Parts

Major Components:

Arm Elbow Piece

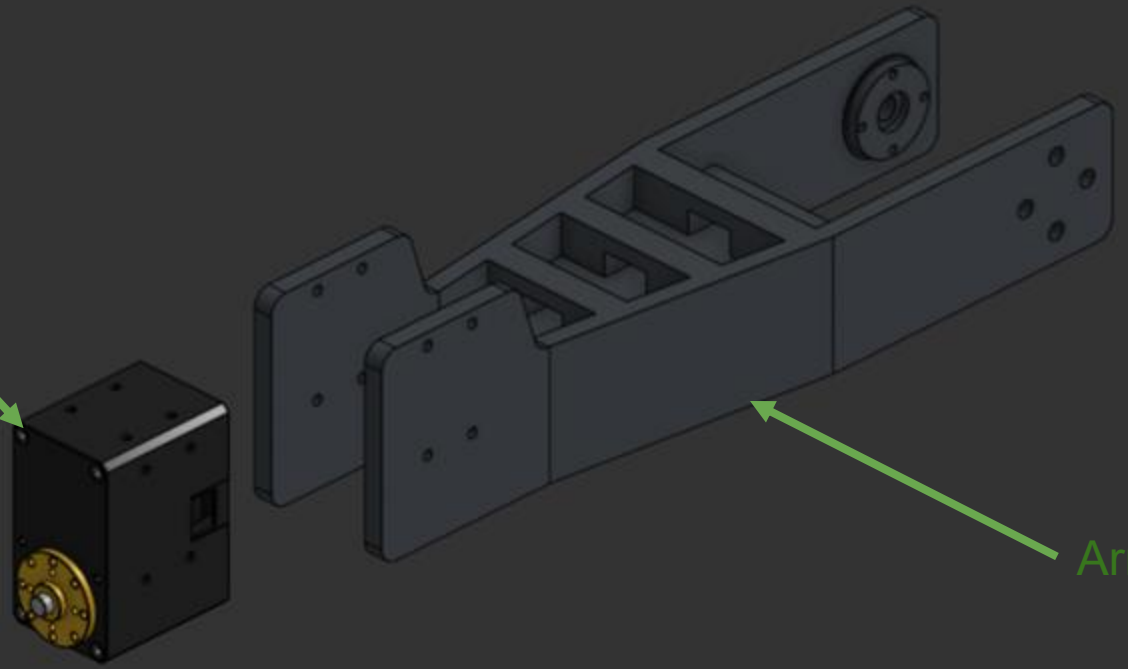
1x Robotis XM540-W270-T Actuator

Minor Components:

8x M2.5x8mm Socket Head Bolt

Major Components

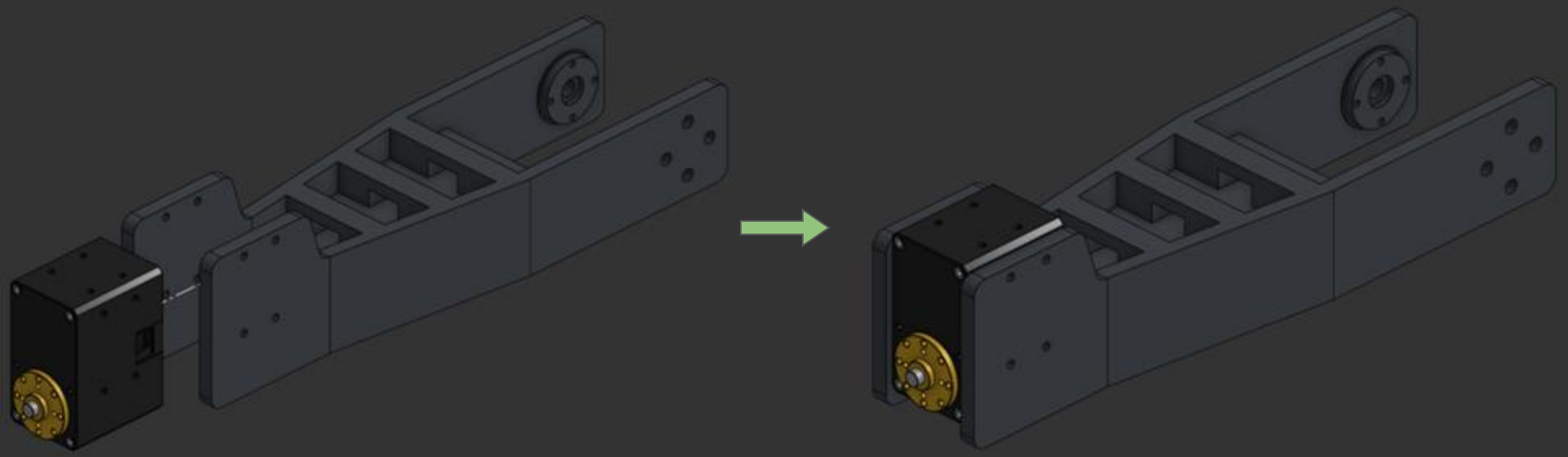
Robotis
XM540-W270-T
Actuator



Arm Elbow Piece

Step 1

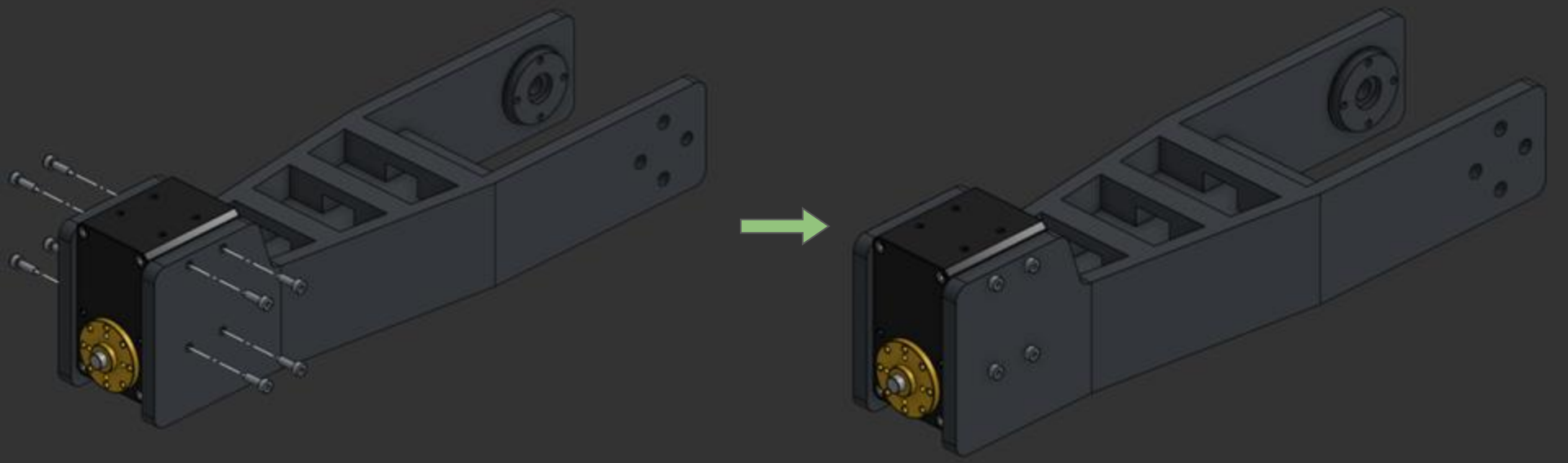
Insert the actuator between the ends of the elbow piece



Parts
-Arm Elbow Piece
-Robotis XM540-W270-T Actuator

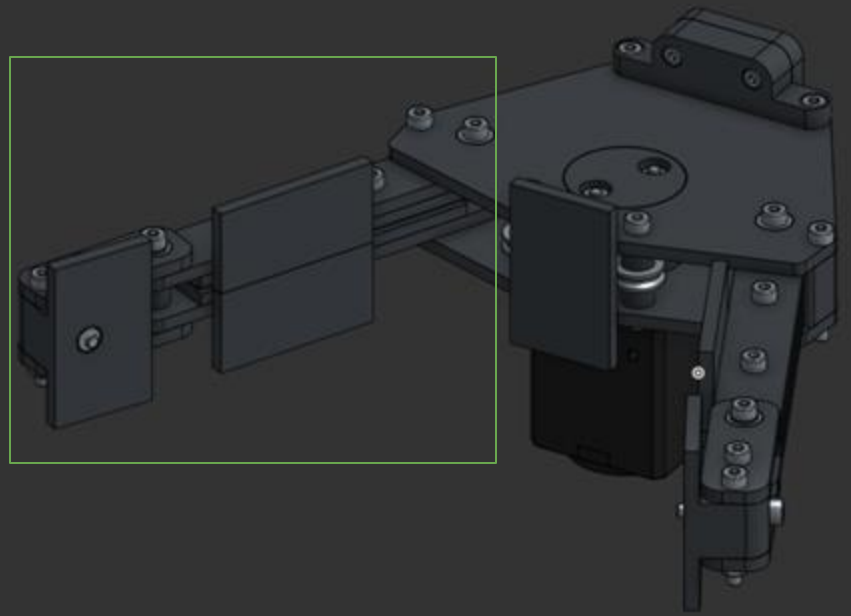
Step 2

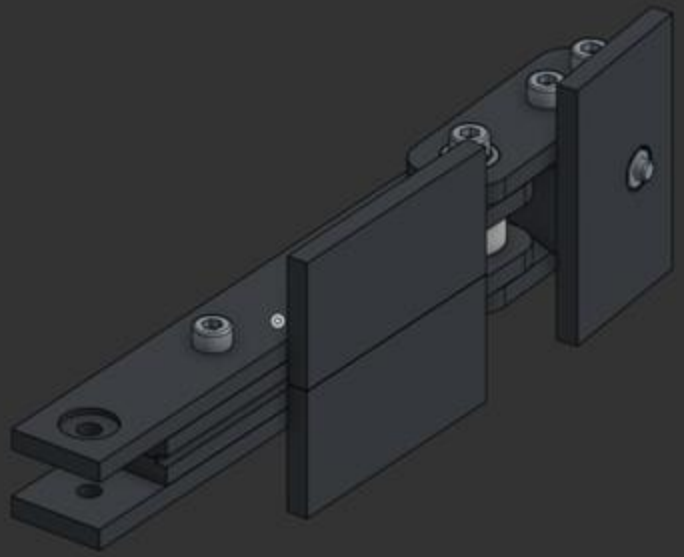
Bolt the actuator into the ends of the elbow piece



Parts
-Previous step
-4x M2.5x8mm Socket Head Bolt

Arm Gripper Finger Assembly Instructions



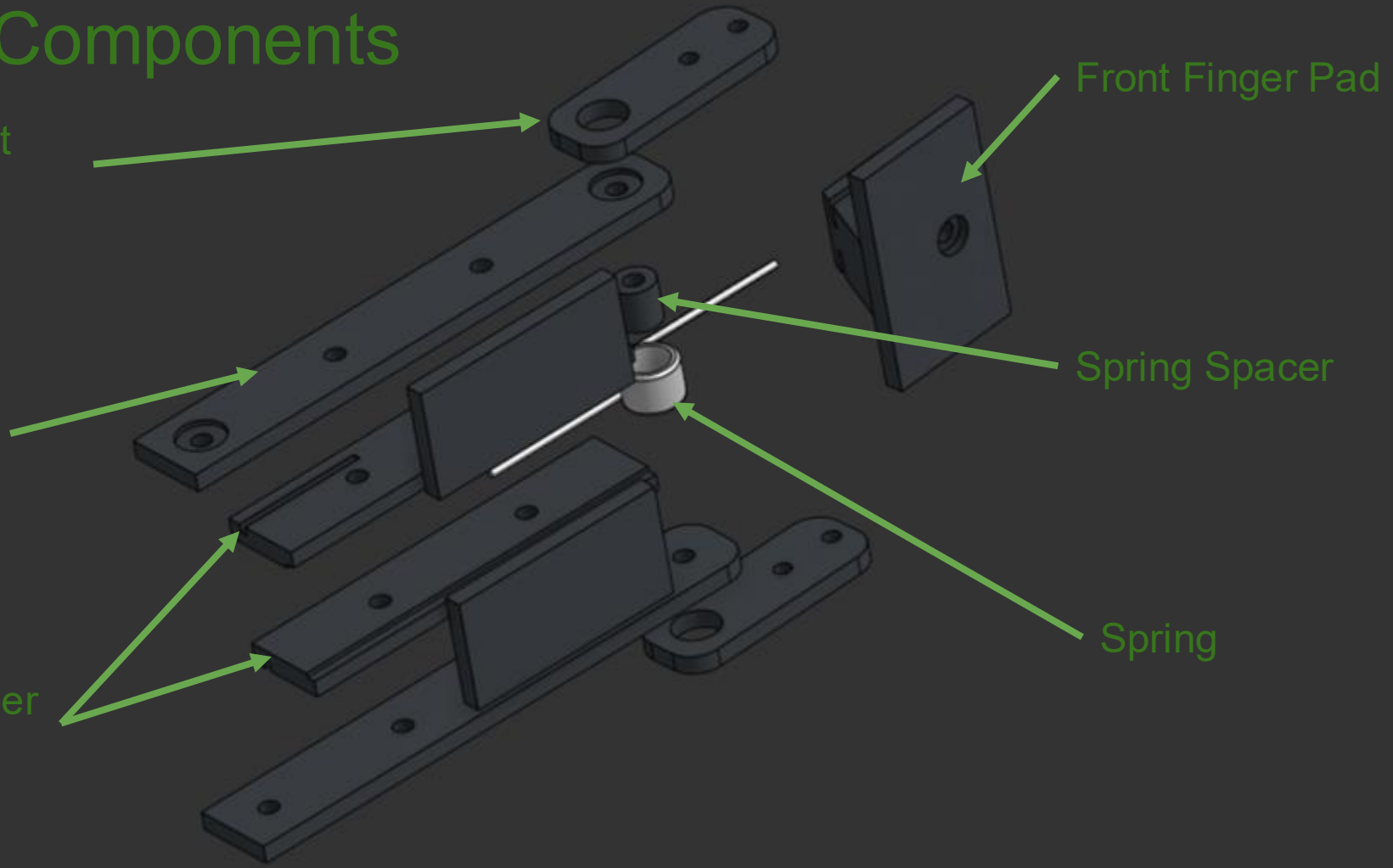


Parts

Major Components:
2x Spring
2x Spring Spacer
4x Finger Front Plate
4x Finger Center Plate
2x Front Finger Pad
4x Center Finger Pad (two top pieces, two bottom pieces)

Minor Components:
6x M3x25mm Socket Head Bolt
4x M3x20mm Socket Head Bolt
4x 3x8x4mm Bearing
10x M3 Lock Washer
10x M3 Nut

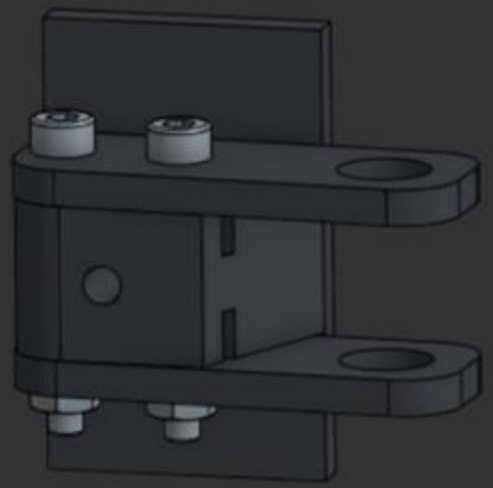
Major Components



Step 1

Bolt the Front Finger Plates to the Front Finger Pad with a lock washer underneath the nut

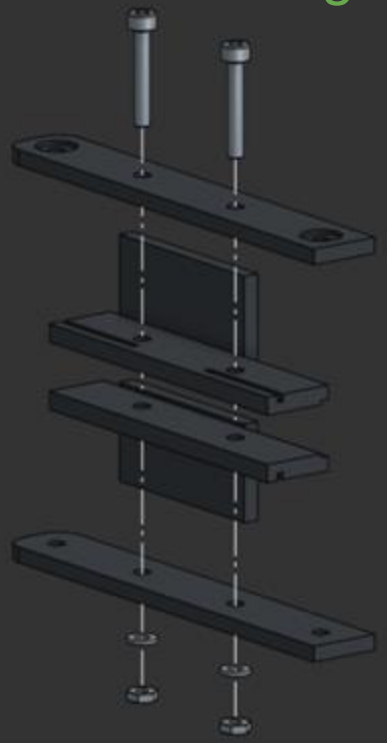




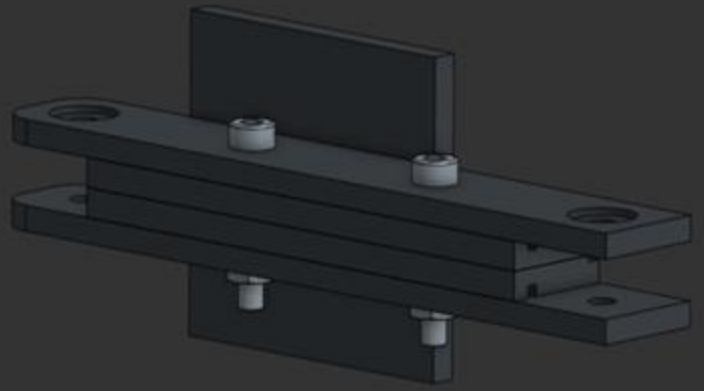
Parts
-2x Finger Front Plate
-Front Finger Pad
-2x M3x25mm Socket Head Bolt
-2x M3 Lock Washer
-2x M3 Nut

Step 2

Bolt the Center Finger Plates and Center Finger Pads together with a lock washer underneath the nut





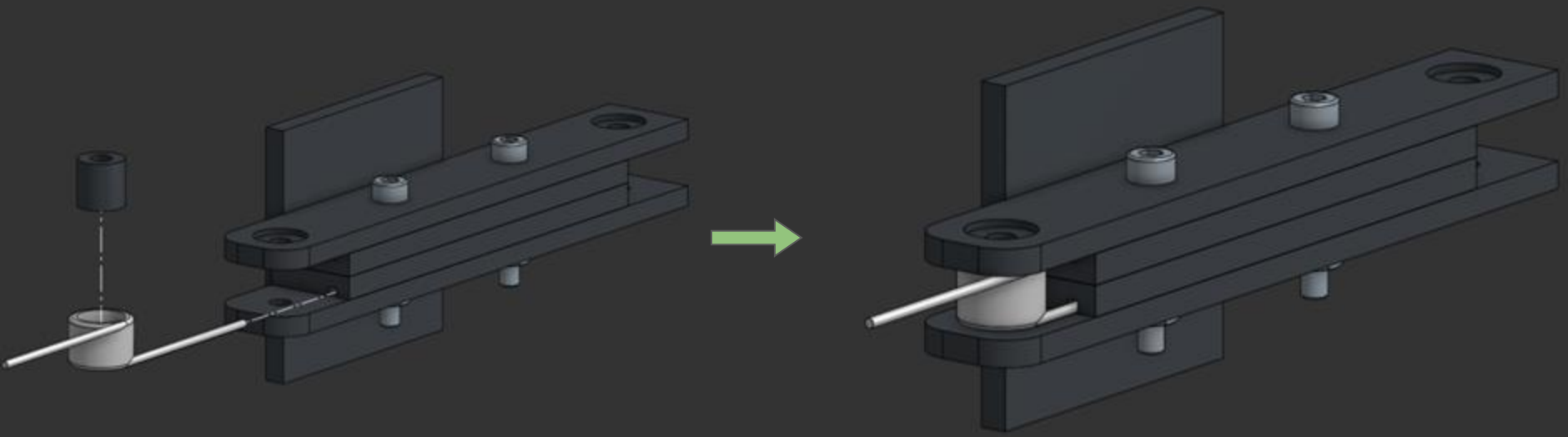


Parts
-2x Finger Center Plate
-2x Center Finger Pad
-2x M3x20mm Socket Head Bolt

-2x M3 Lock Washer
-2x M3 Nut

Step 3

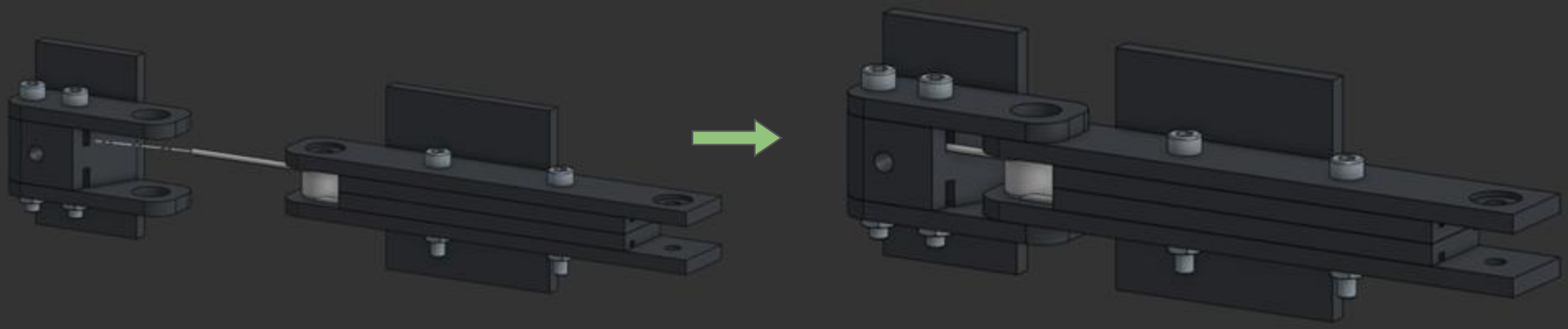
Insert the Spring Spacer into the center of the spring and then align the spring with the end of the Center Finger Plates



Parts
-Previous step
-Spring
-Spring Spacer

Step 4

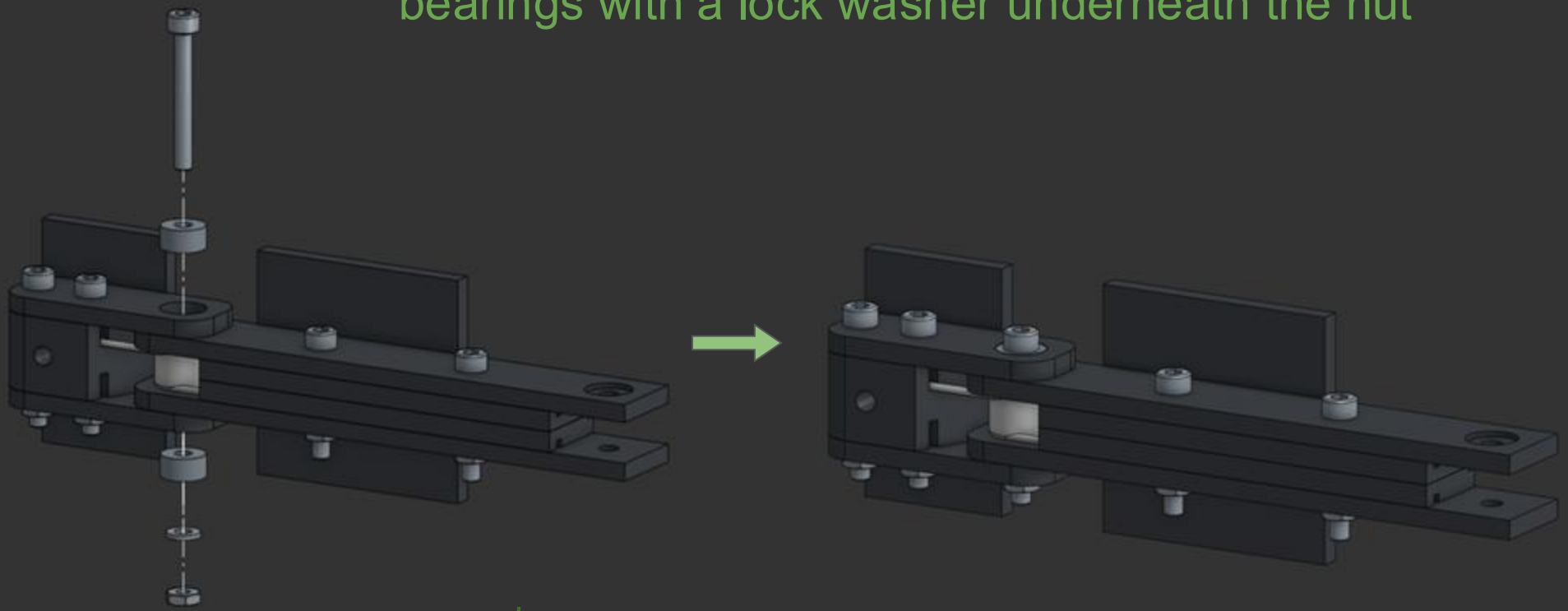
Align the front section of the finger with the center section



Parts
-Previous step
-Finger Front Assembly

Step 5

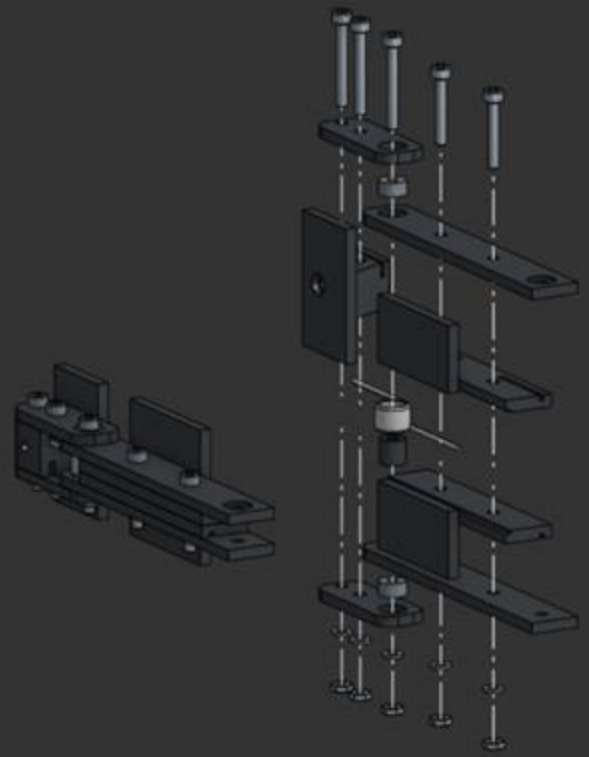
Bolt the front and center sections together on top of the bearings with a lock washer underneath the nut



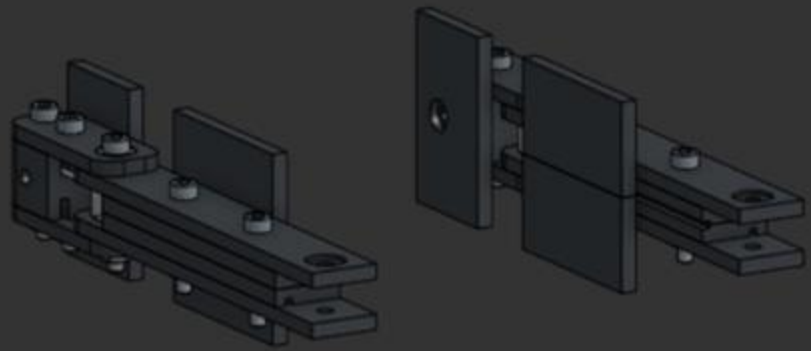
Parts
-Previous step
-M3x25mm Socket Head Bolt
-2x 3x8x4mm Bearing
-M3 Lock Washer
-M3 Nut

Step 6

Repeat steps 1-5 for a mirror finger





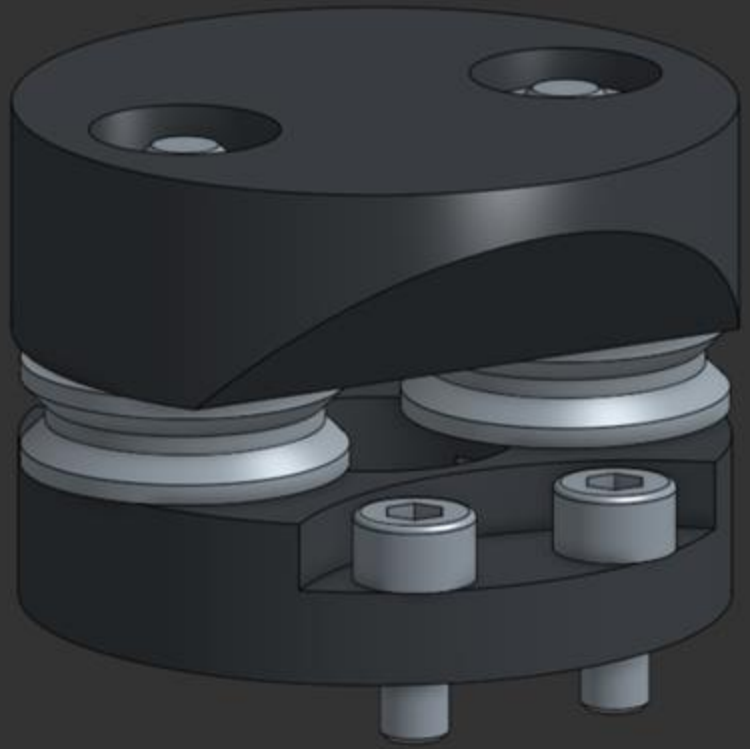


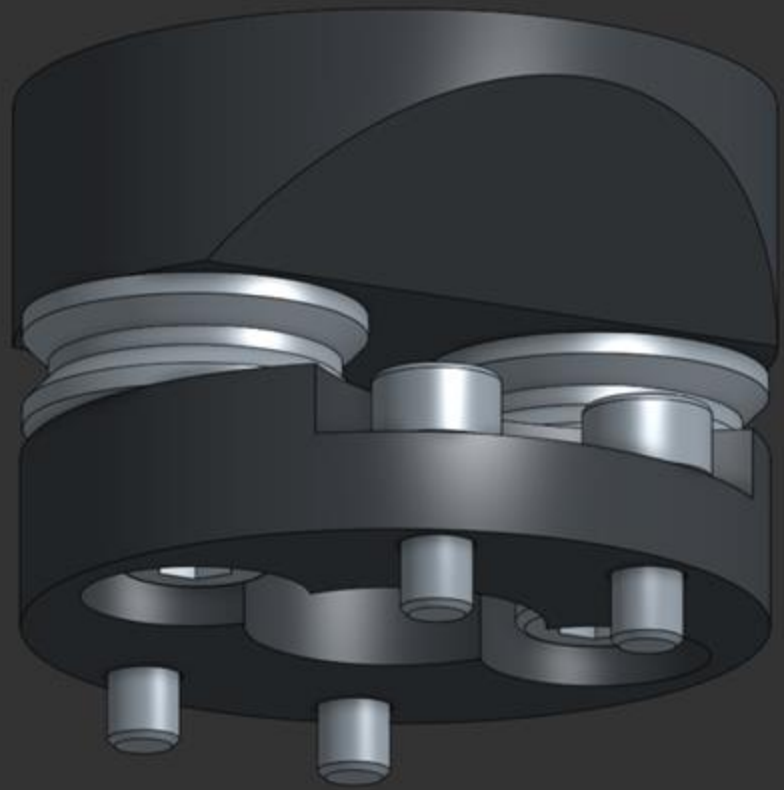
Parts
-Previous step
-Front Finger Pad
-2x Finger Front Plate

-2x Finger Center Pads
-2x Finger Center Plate
-Spring
-Spring Spacer

-2x 3x8x4mm Bearing
-2x M3x20mm Socket Head Bolt
-3x M3x25mm Socket Head Bolt
-2x M3 Lock Washer
-2x M3 Nut

Arm Gripper Coiler Assembly Instructions



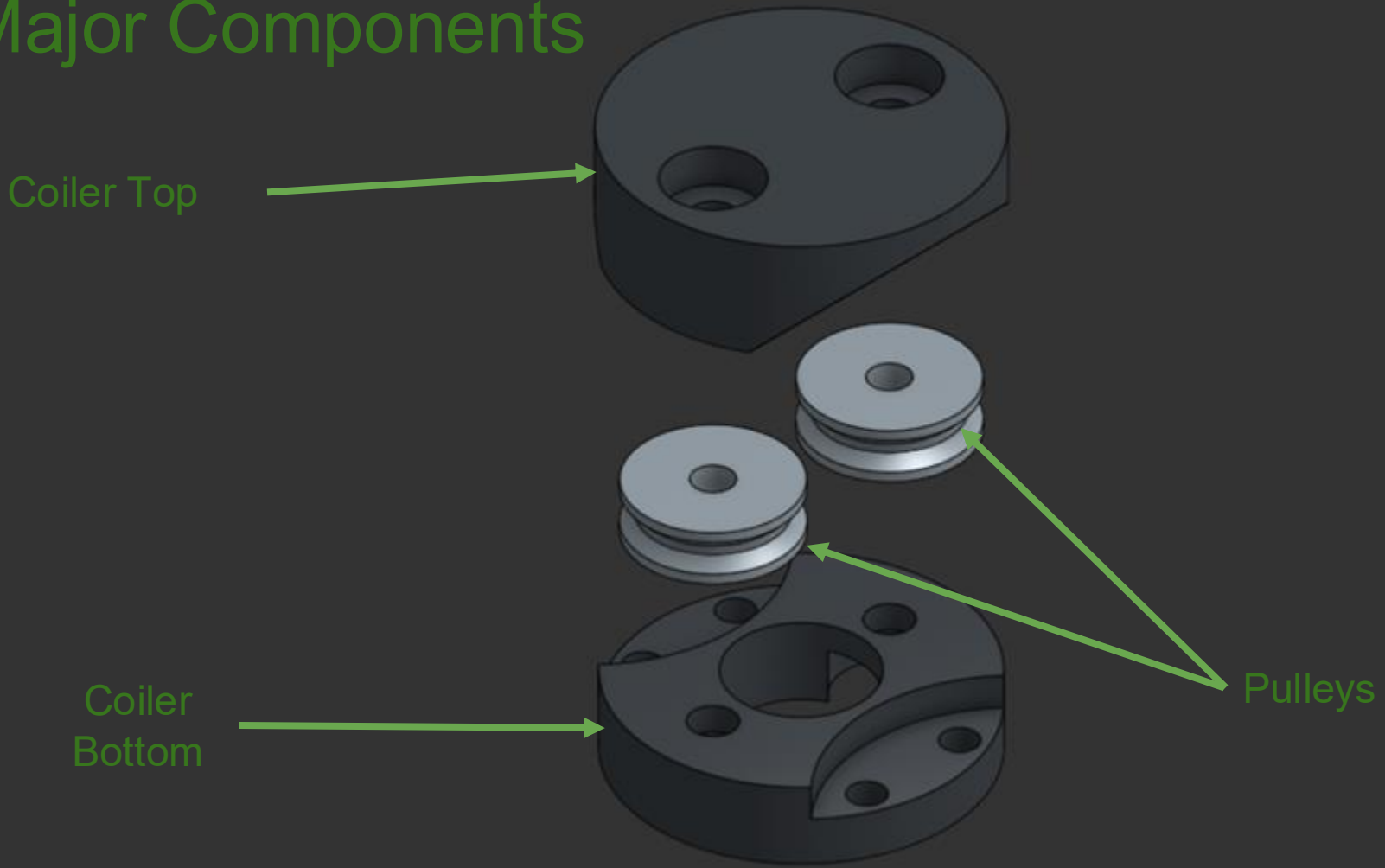


Parts

Major Components:
2x Spring
2x Spring Spacer
4x Finger Front Plate
4x Finger Center Plate
2x Front Finger Pad
4x Center Finger Pad (two top pieces, two bottom pieces)

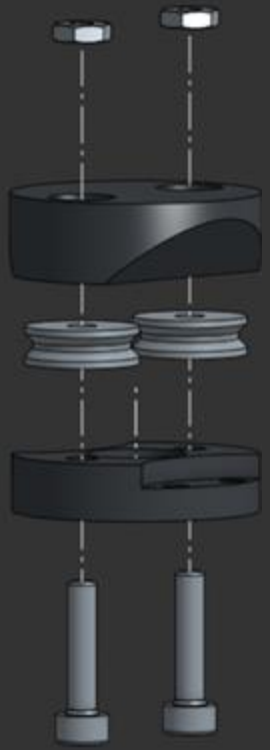
Minor Components:
8x M2.5x8mm Bolt

Major Components

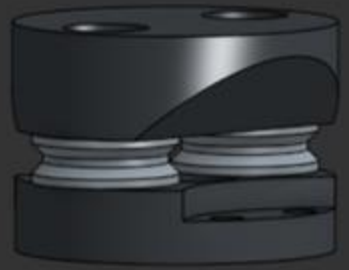


Step 1

Bolt the Coiler Top to the pulleys and Coiler Bottom





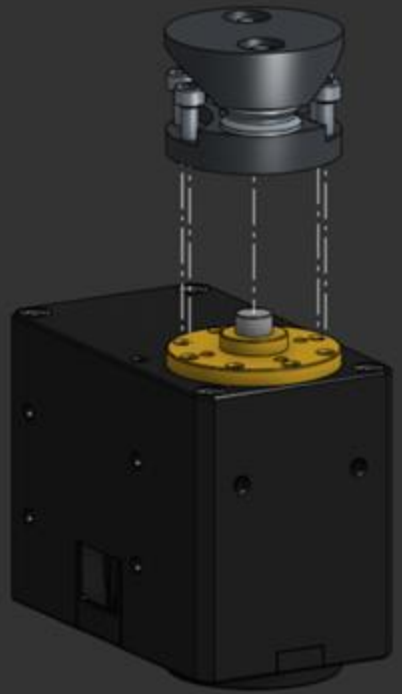


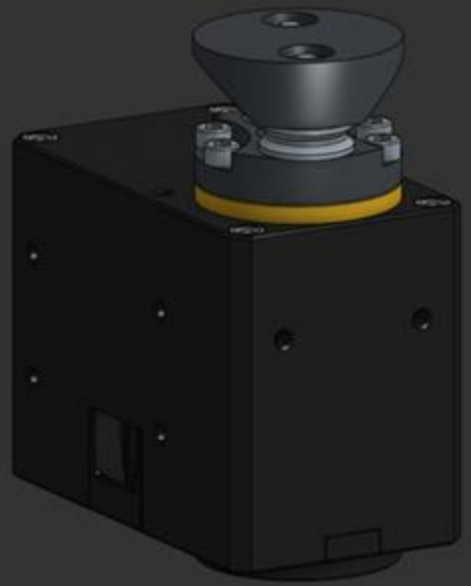
Parts
-Coiler Top
-Coiler Bottom
-2x Pulley

-2x M3x14mm Socket Head Bolt
-2x M3 Nut

Step 2

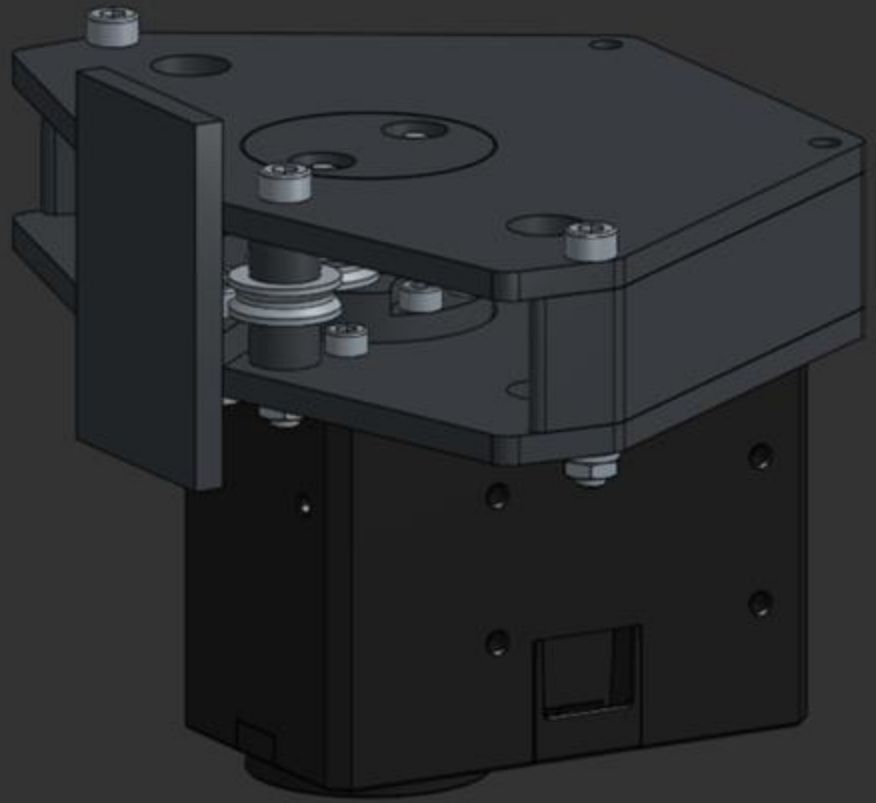
Bolt the Coiler to the horn of the actuator

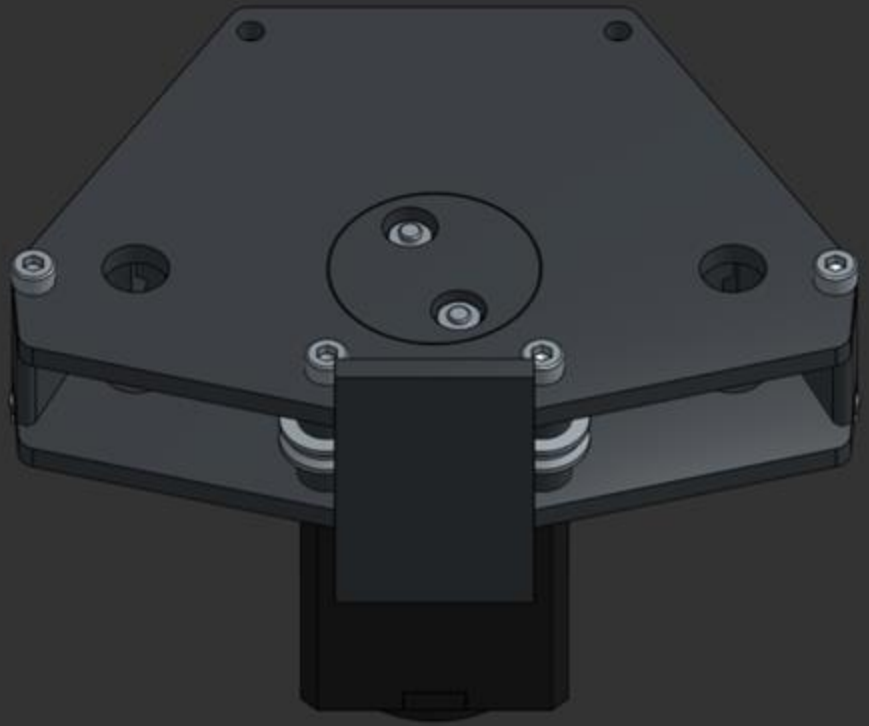




Parts
-Previous step
-Robotis XM540-W270-T Actuator
-4x M2.5x8mm Socket Head Bolt

Arm Gripper Base Assembly Instructions



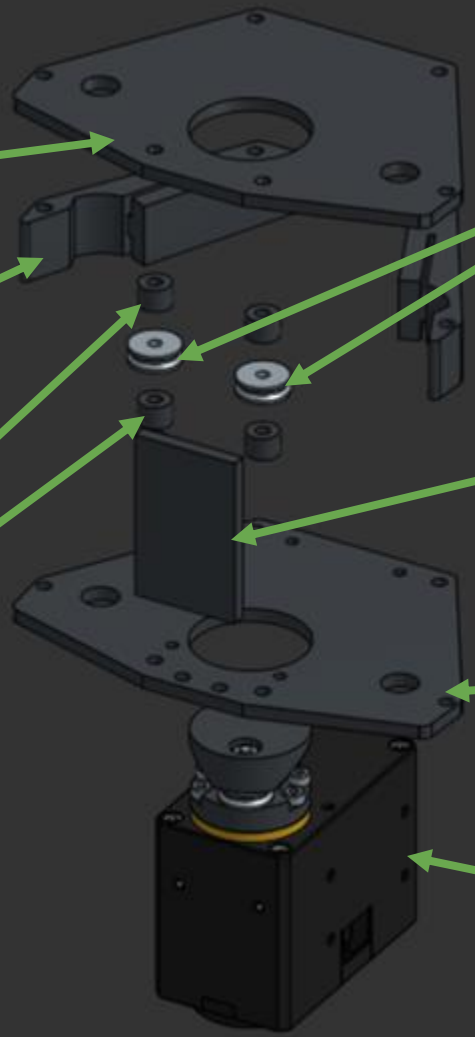


Parts

Major Components:
Coiler Assembly
Gripper Base Top Plate
Gripper Base Bottom Plate
Gripper Base Front
2x Gripper Base Side Wall
2x Pulley
2x Pulley Spacer

Minor Components:
2x M3x10mm Socket Head Bolt
4x M3x25mm Socket Head Bolt
6x M3 Lock Washer
6x M3 Nut

Major Components



Gripper Base Top Plate

Pulleys

Gripper Base Side Wall

Gripper Base Front

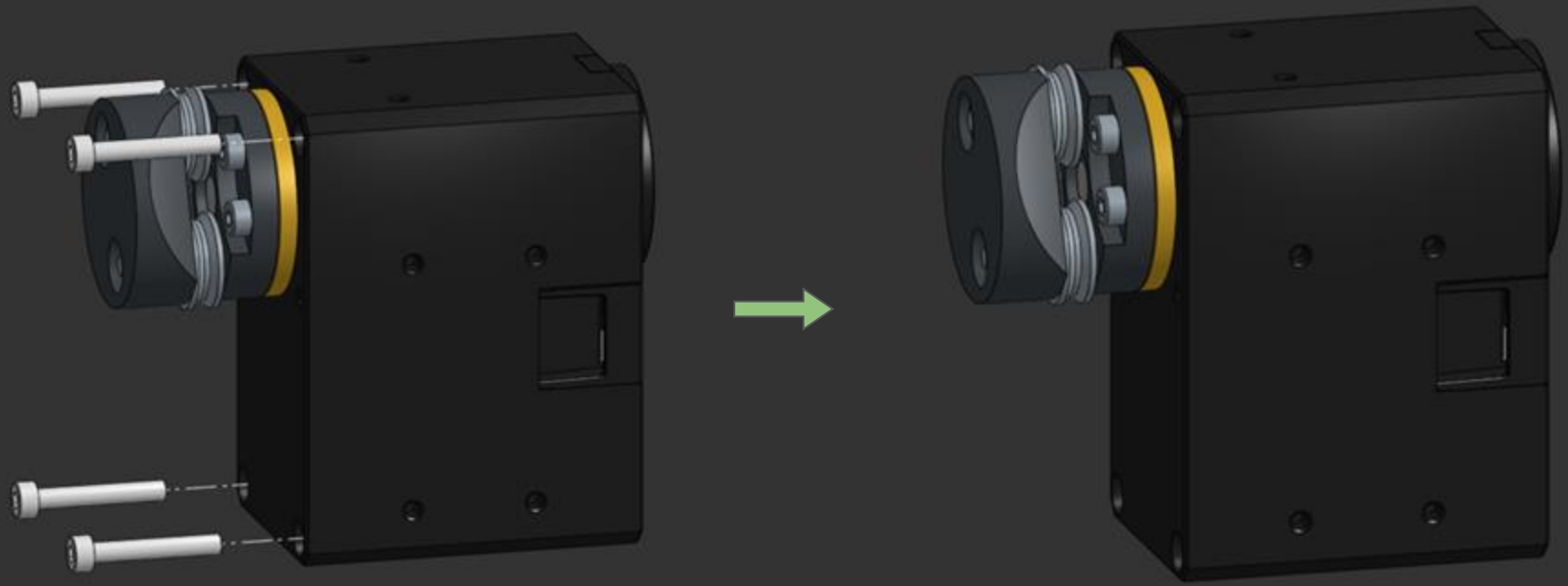
Gripper Base Bottom Plate

Pulley Spacers

Coiler Assembly

Step 1

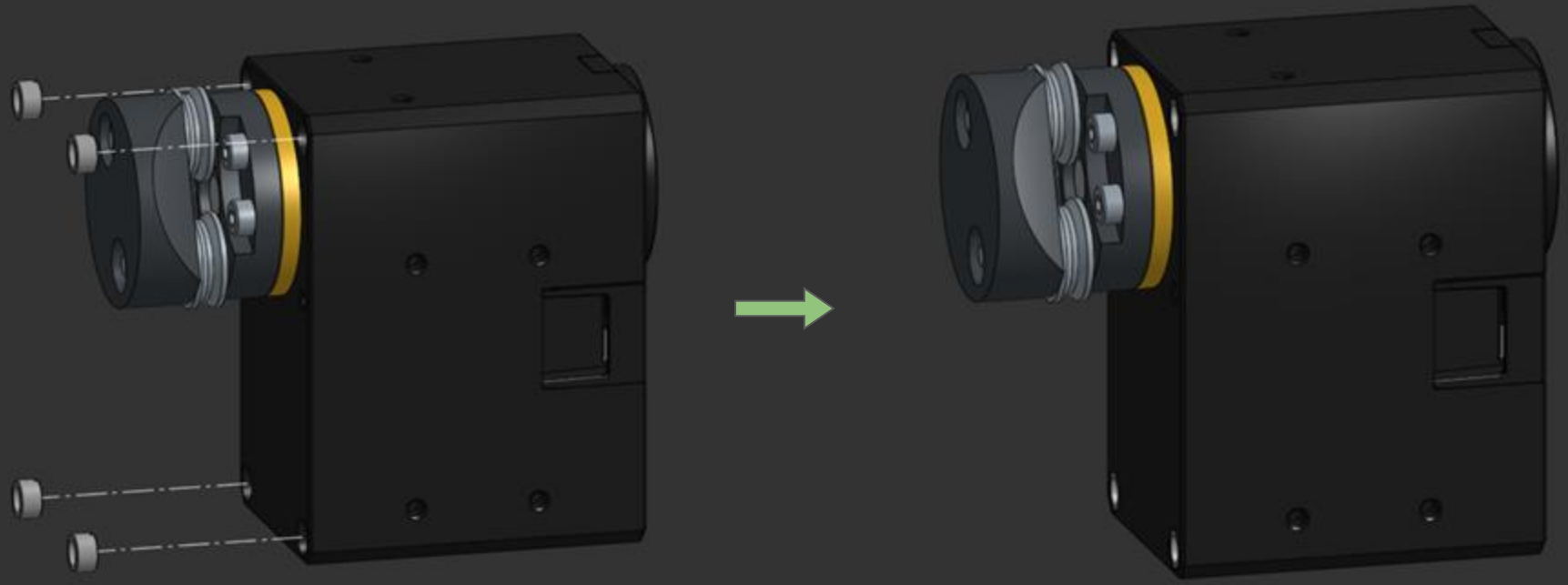
Remove the front four bolts from the actuator



Parts
-Coiler Assembly

Step 2

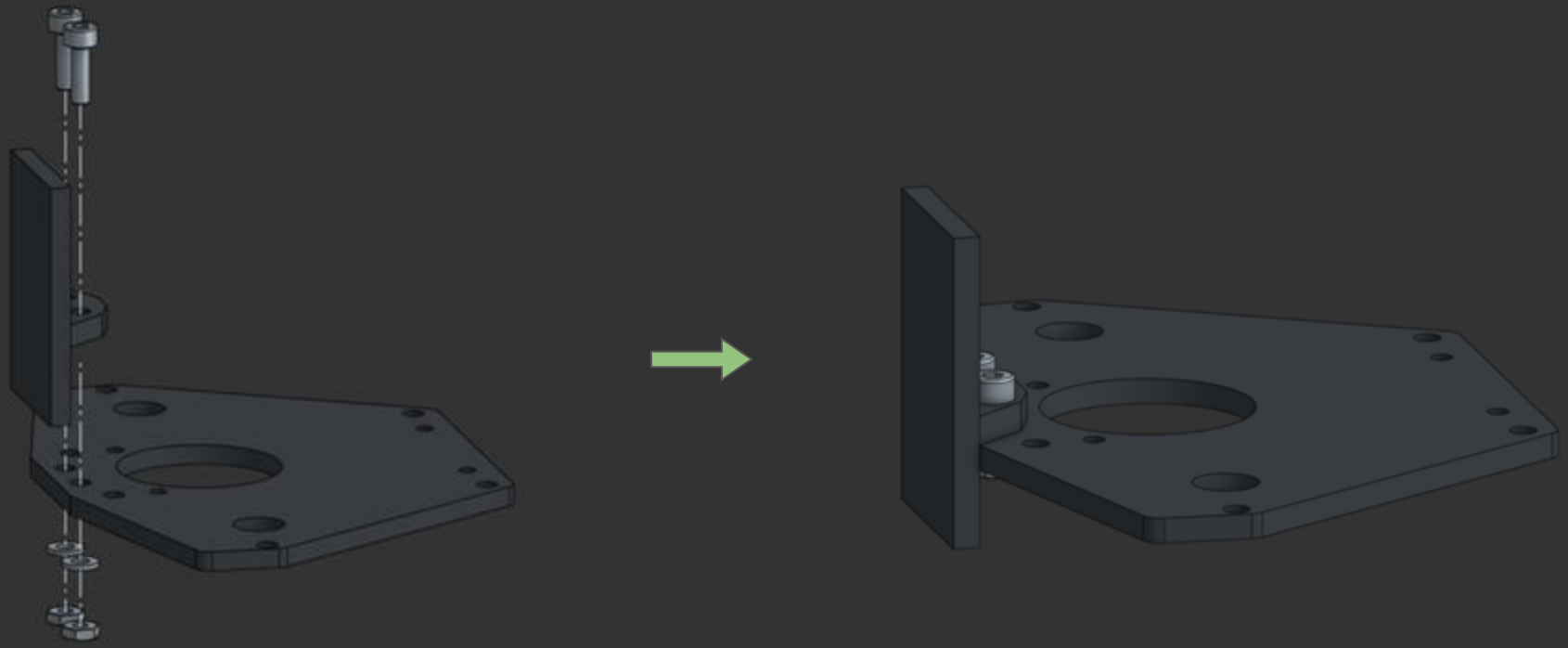
Insert Robotis Spacers into the front of the actuator



Parts
-Previous step
-4 Robotis Spacers

Step 3

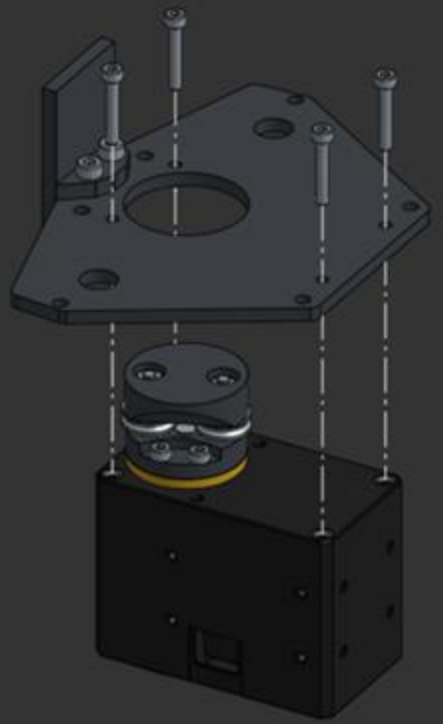
Bolt the Front Finger Plates to the Front Finger Pad with a lock washer underneath the nut

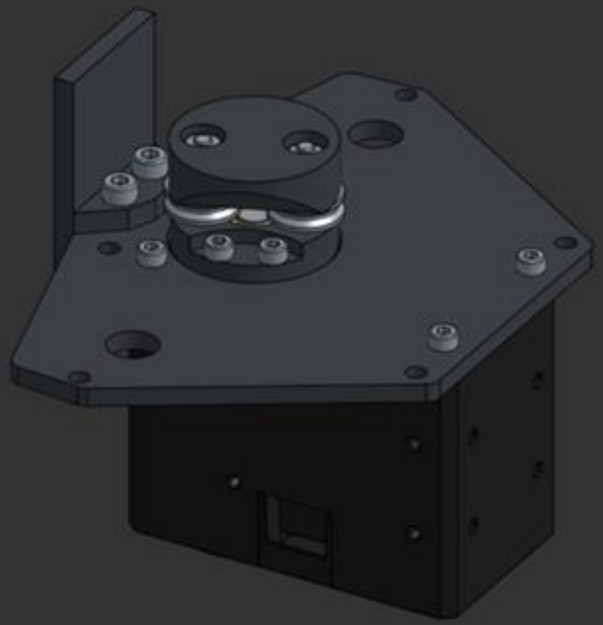


Parts
-Gripper Base Bottom Plate
-Gripper Base Front
-2x M3x10mm Socket Head Bolt
-2x M3 Lock Washer

Step 4

Bolt the Gripper Base Bottom Plate to the actuator

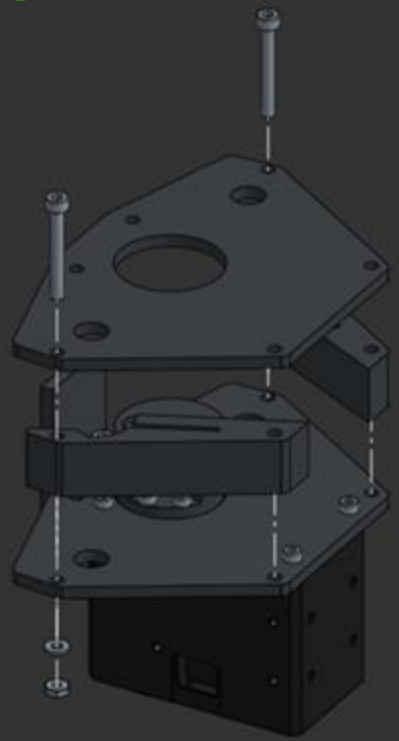


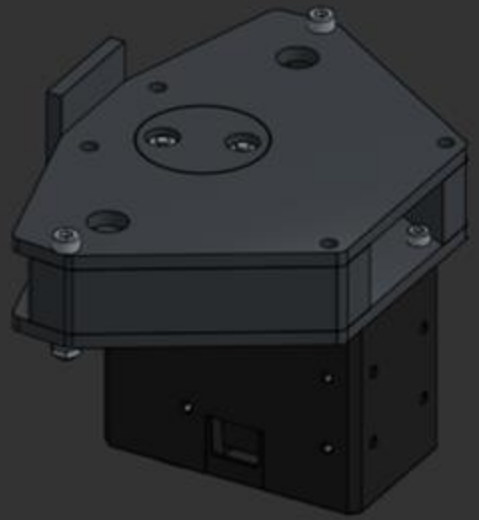


Parts
-Previous step
-Coiler Assembly with spacers
-4x M2.5x16mm Socket Head Bolt

Step 5

Bolt the Front Finger Plates to the Front Finger Pad with a lock washer underneath the nut



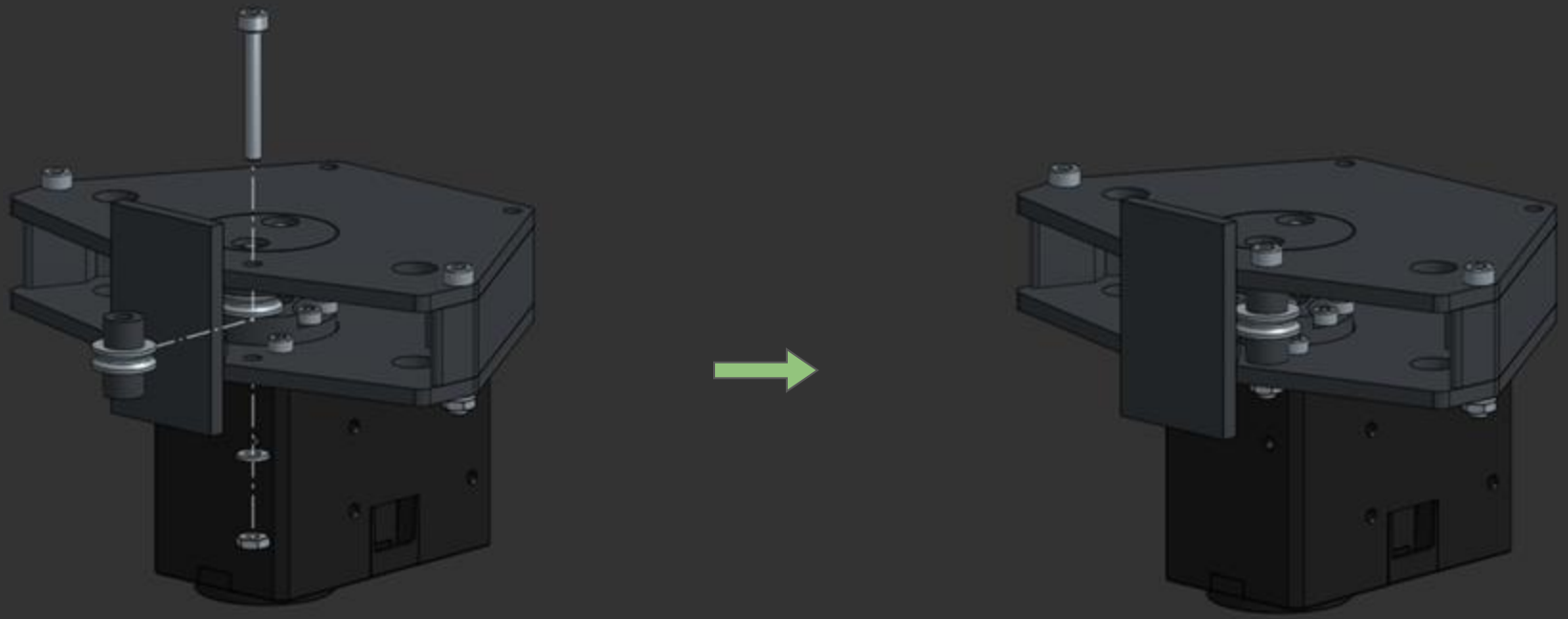


Parts
-Previous step
-Gripper Base Side Wall
-2x M3x25mm Socket Head Bolt

-2x Lock Washer
-2x M3 Nut

Step 6

Bolt the pulley and pulley spacers to the front of the Gripper Base

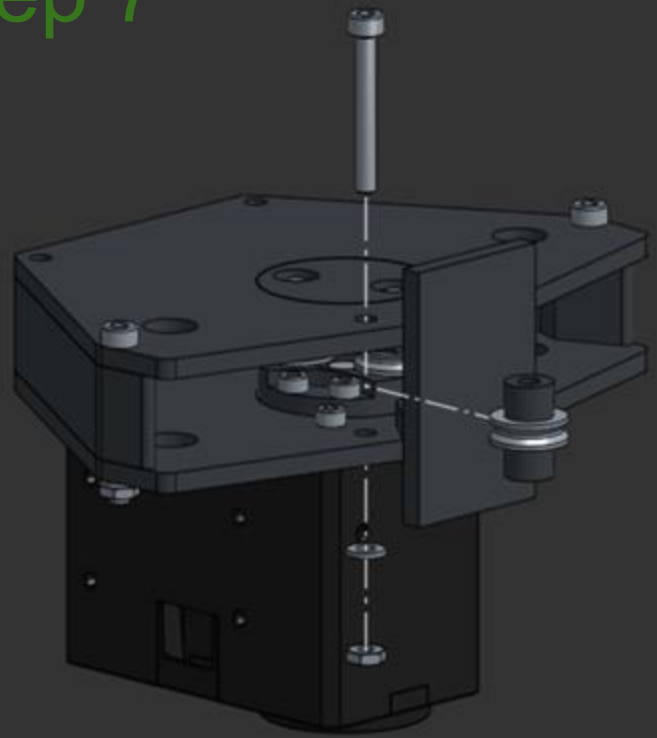


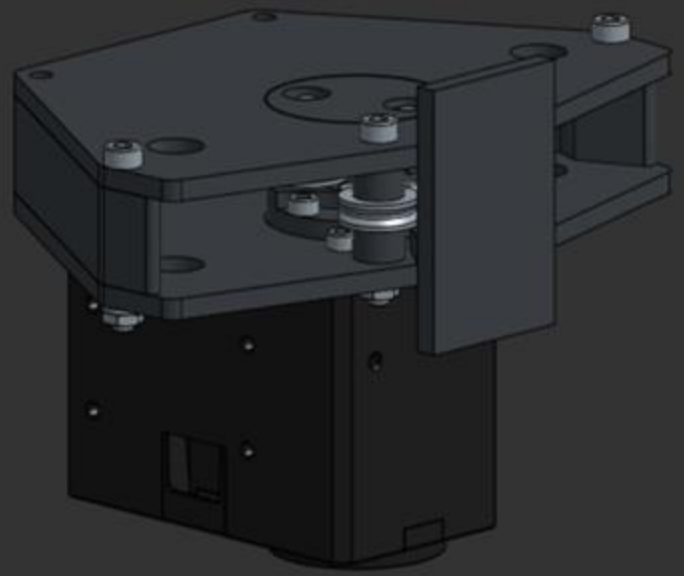
Parts
-Previous step
-3x12x4mm Pulley
-2x Pulley Spacer

-M3x25mm Socket Head Bolt
-M3 Lock Washer
-M3 Nut

Step 7

Repeat for the other side

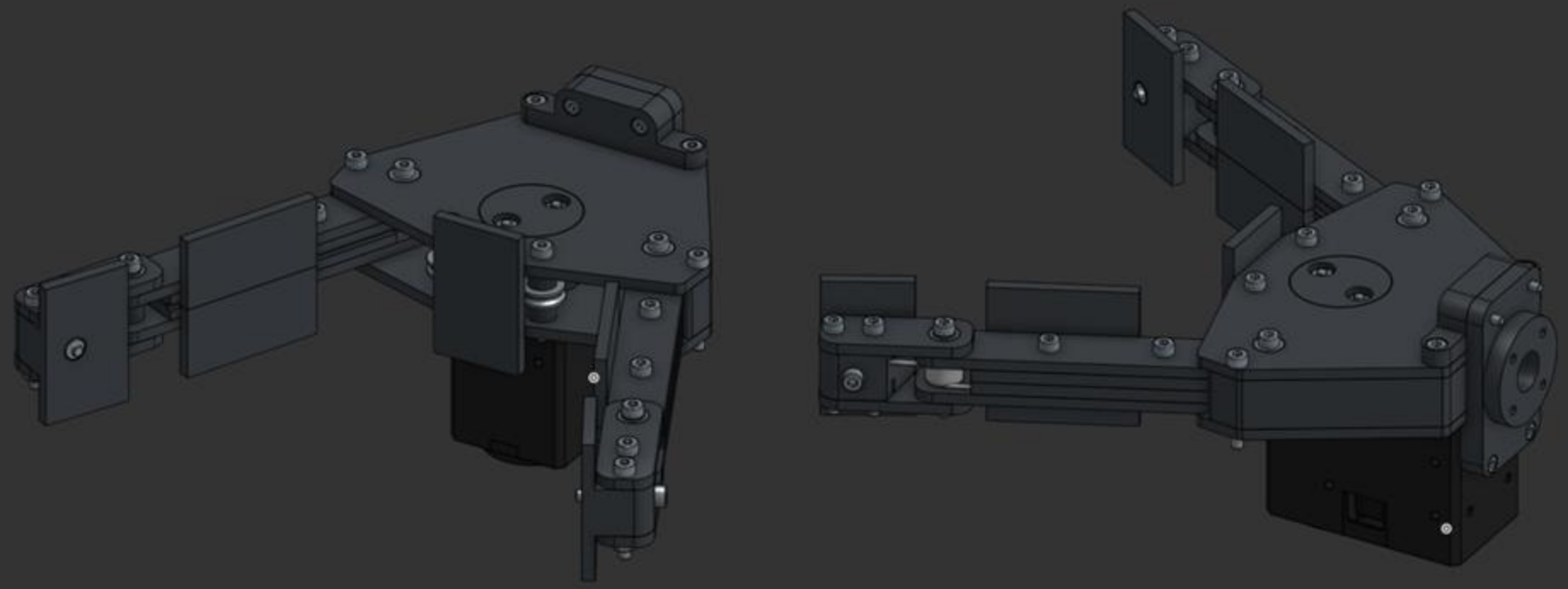




Parts
-Previous step
-3x12x4mm Pulley
-2x Pulley Spacer

-M3x25mm Socket Head Bolt
-M3 Lock Washer
-M3 Nut

Arm Gripper Assembly Instructions



Parts

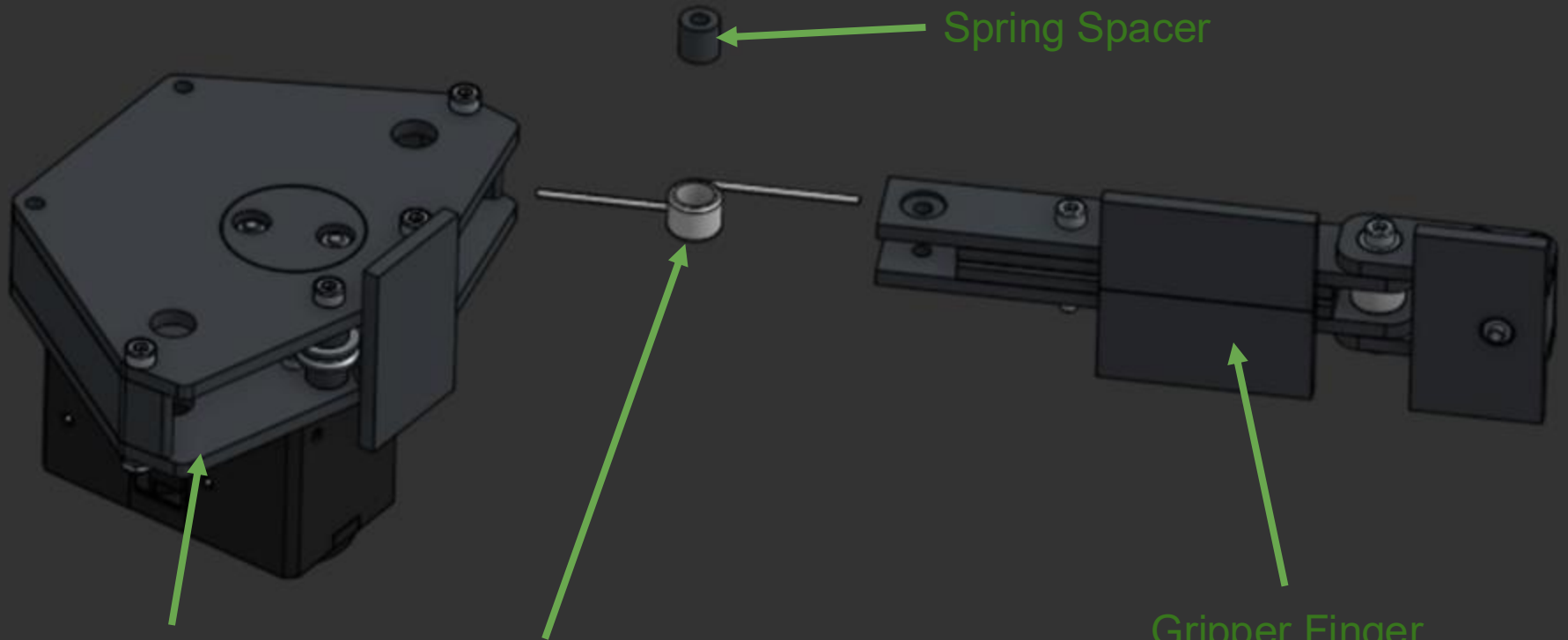
Major Components:

Gripper Base Assembly
2x Gripper Finger Assembly
2x Spring
2x Spring Spacer

Minor Components:

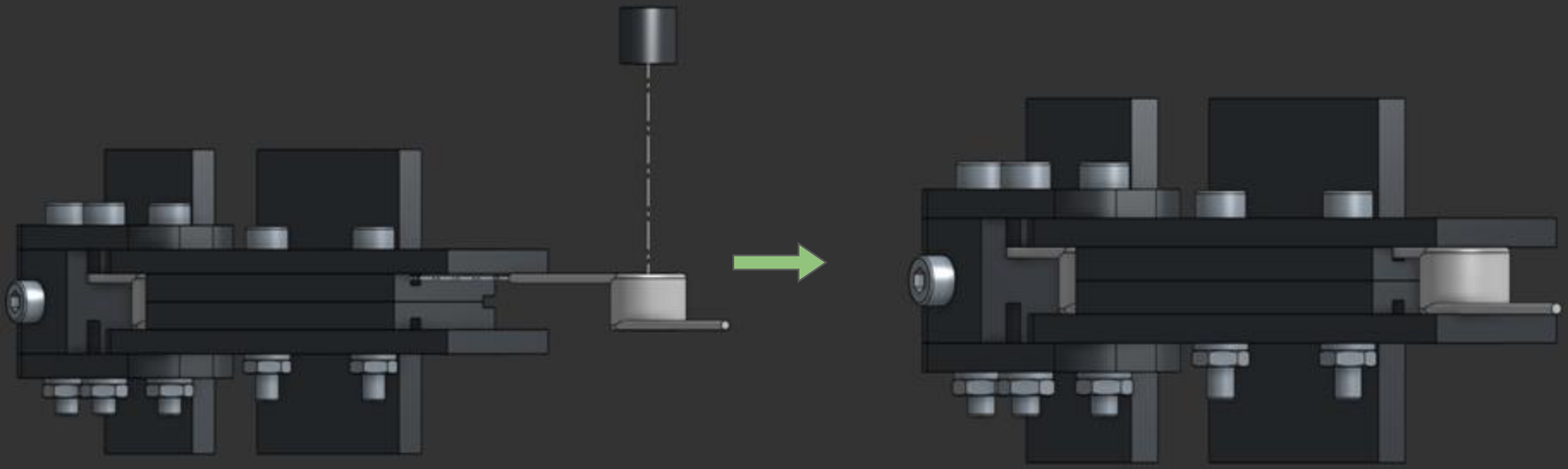
2x M3x25mm Socket Head Bolt
2x M3 Lock Washer
2x M3 Nut

Major Components



Step 1

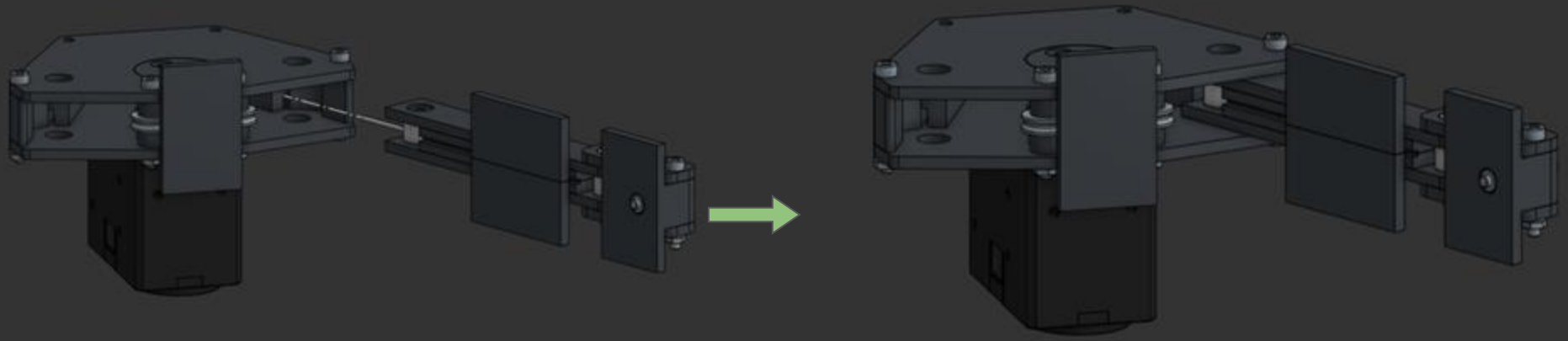
Insert a spring spacer into the spring and align the spring with the end of the finger assembly



Parts
-Gripper Finger Assembly
-Spring
-Spring Spacer

Step 2

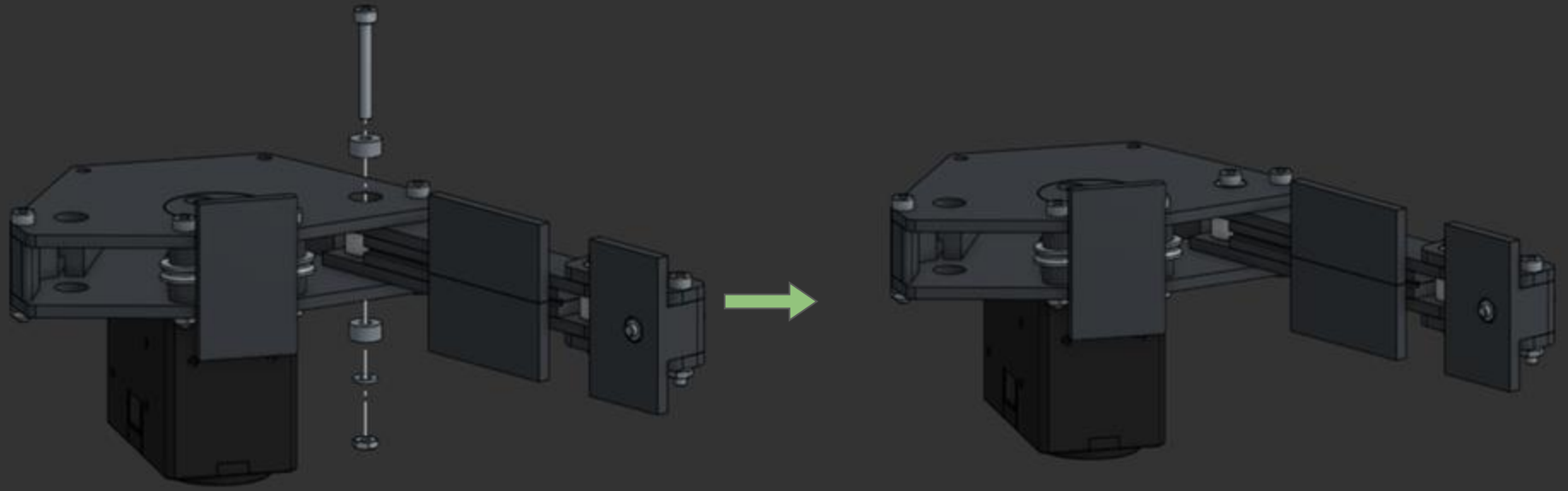
Insert the end of the spring into the Gripper Base Side Wall



Parts
-Previous step
-Gripper Base Assembly

Step 3

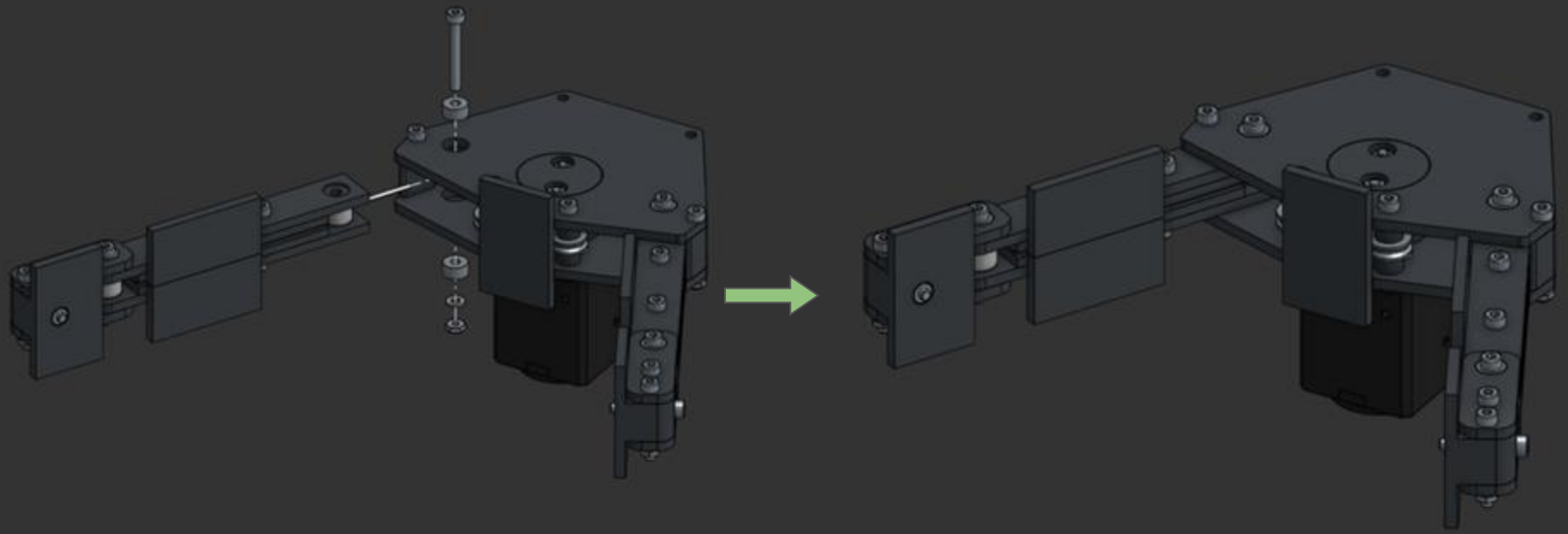
Bolt the finger and two bearings to the gripper base with a lock washer and nut underneath



Parts
-Previous step
-M3x25mm Socket Head Bolt
-2x 3x8x4mm Bearing
-M3 Lock Washer
-M3 Nut

Step 4

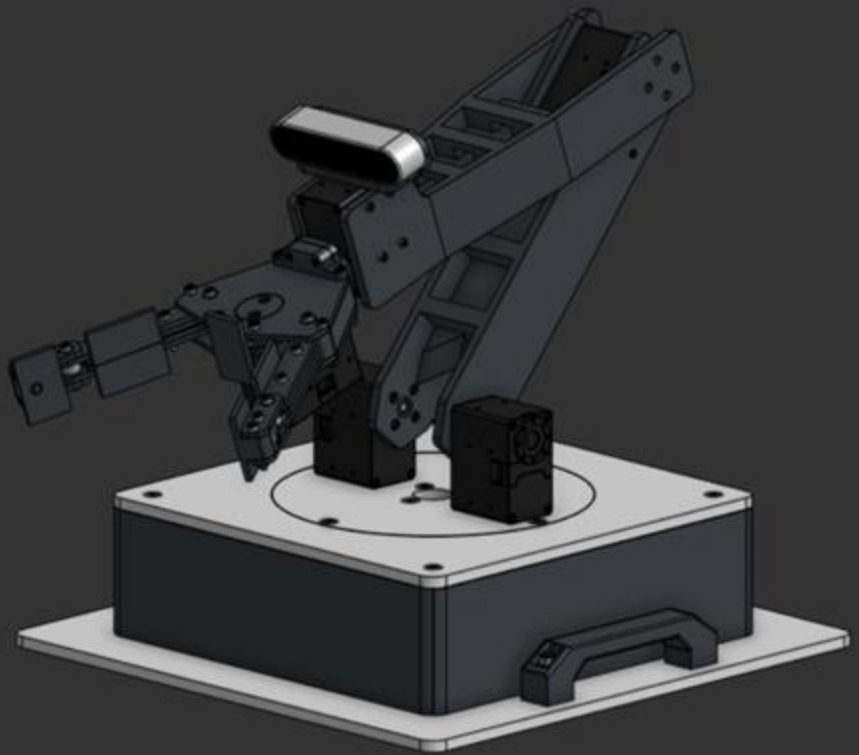
Repeat steps 1-3 for the other finger

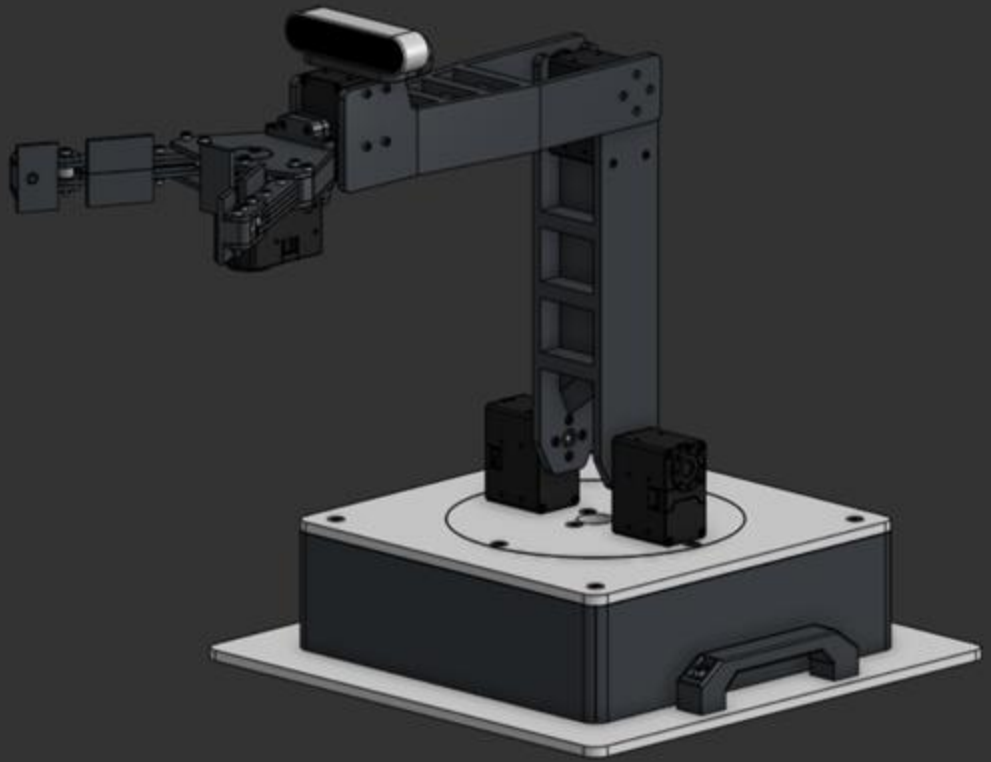


Parts
-Previous step
-Gripper Finger Assembly
-M3x25mm Socket Head Bolt

-2x 3x8x4mm Bearing
-M3 Lock Washer
-M3 Nut

Arm Assembly Instructions





Parts

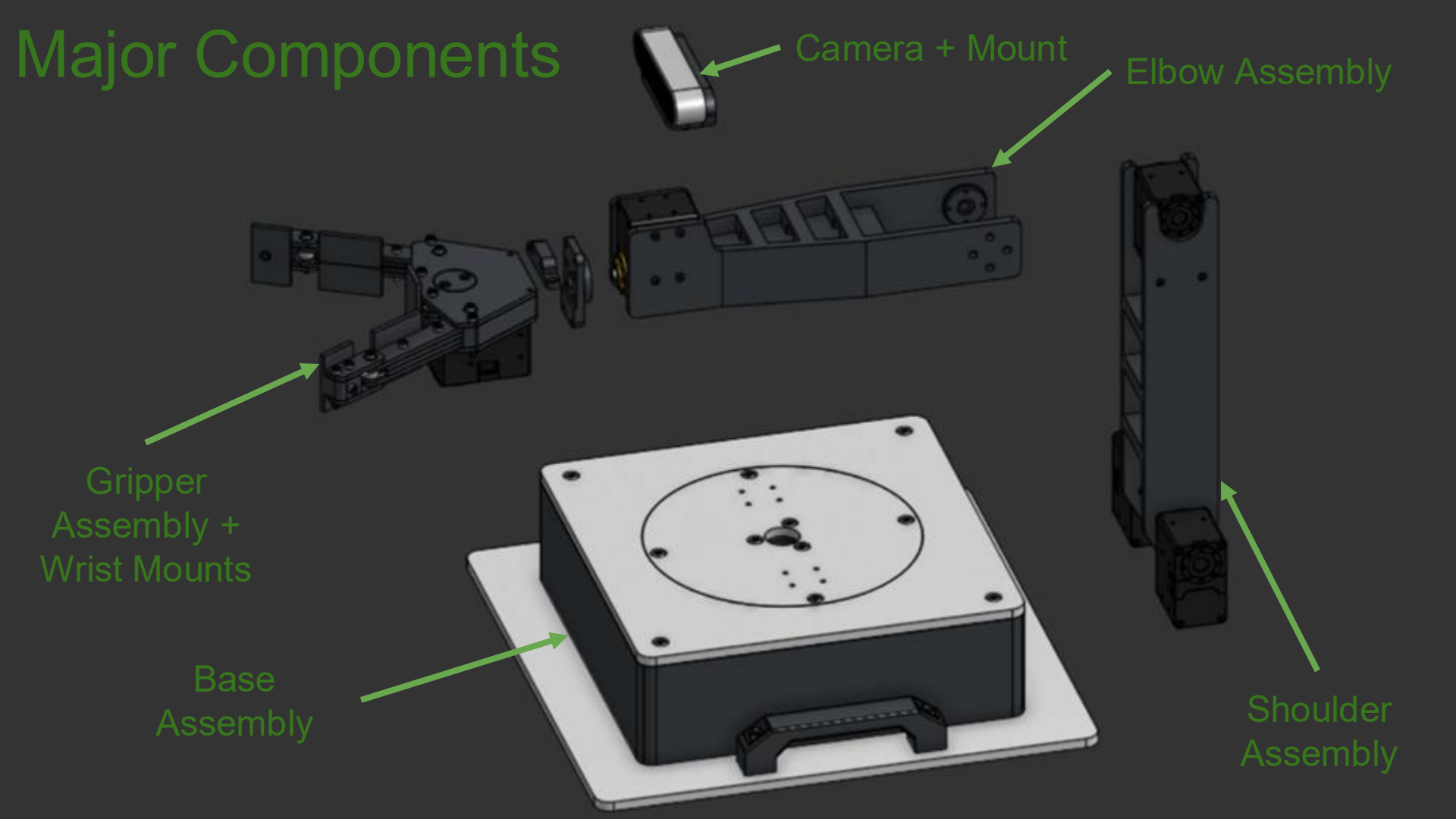
Major Components:

Gripper Assembly
Base Assembly
Shoulder Assembly
Elbow Assembly
Intel Realsense D435i Camera
Camera Mount
Upper Wrist Mount
Center Wrist Mount

Minor Components:

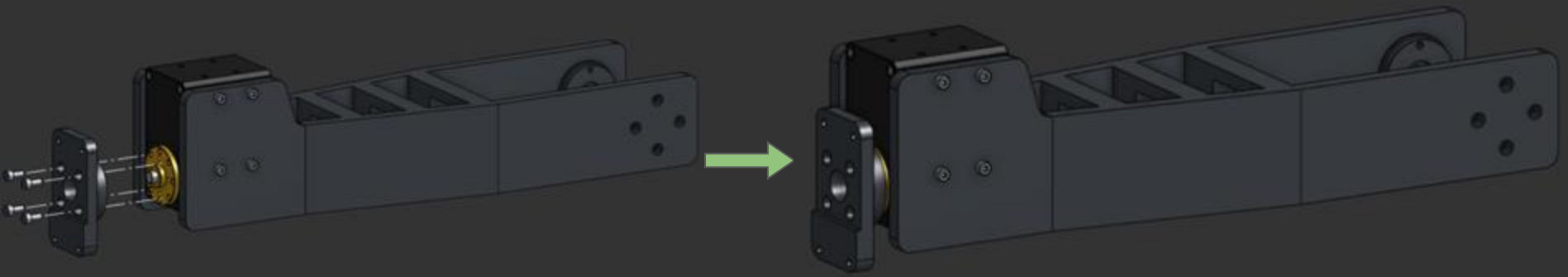
14x M2.5x6mm Socket Head Bolt
10x M2.5x8mm Socket Head Bolt
2x M2.5x14mm Socket Head Bolt
2x M3x25mm Socket Head Bolt
2x M3x10mm Socket Head Bolt
2x M3 Lock Washer
2x M3 Nut
2x M2.5 Lock Washer
2x M2.5 Nut

Major Components



Step 1

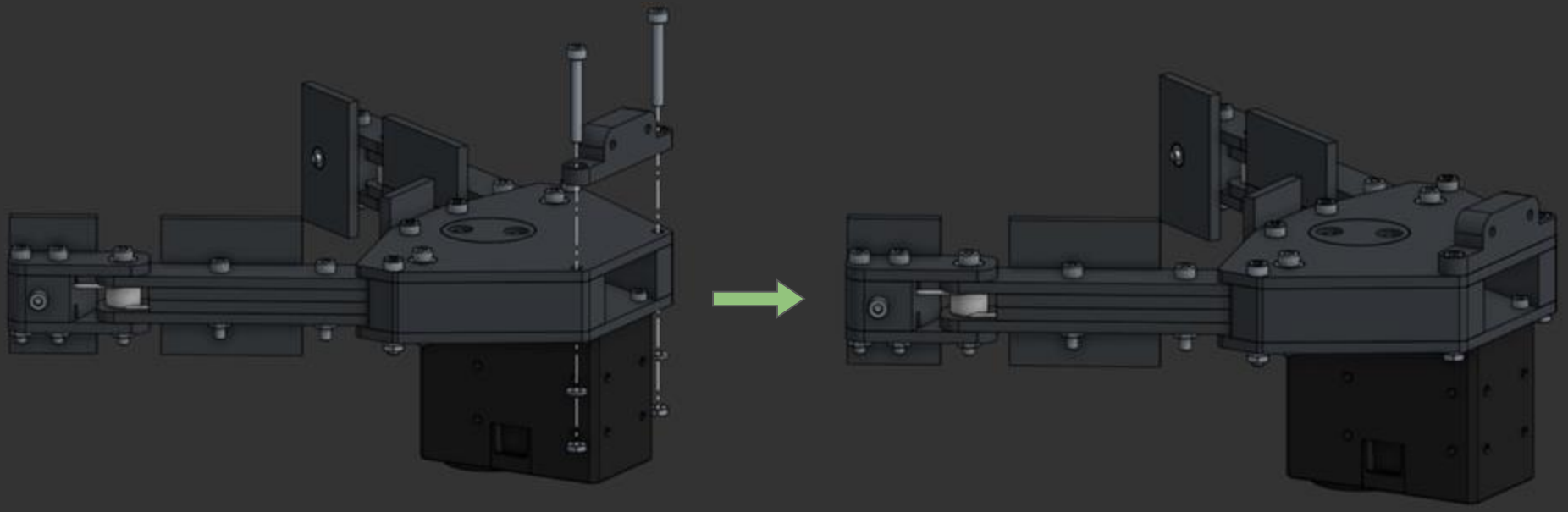
Attach the Center Wrist mount to the Elbow Assembly using 4 bolts



Parts
-Elbow Assembly
-Center Wrist Mount
-4x M2.5x6mm Socket Head Bolts

Step 2

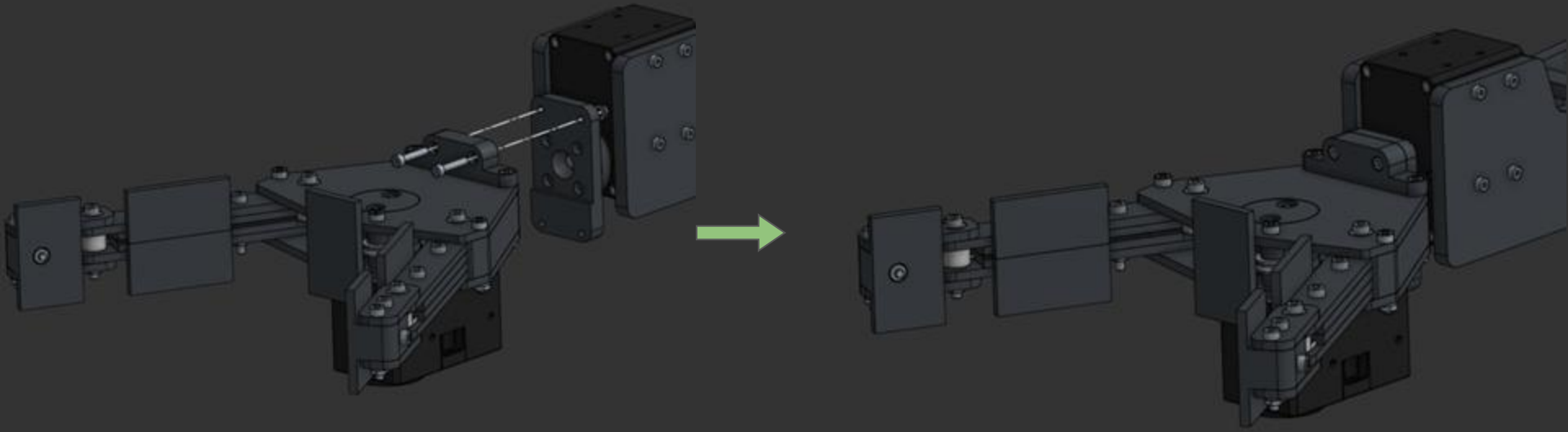
Bolt the Upper Wrist Mount to the Gripper Assembly



Parts
-Top Wrist Mount
-Gripper Assembly
-2x M3x25mm Socket Head Bolts
-2x M3 Lock Washer
-2x M3 Nut

Step 3

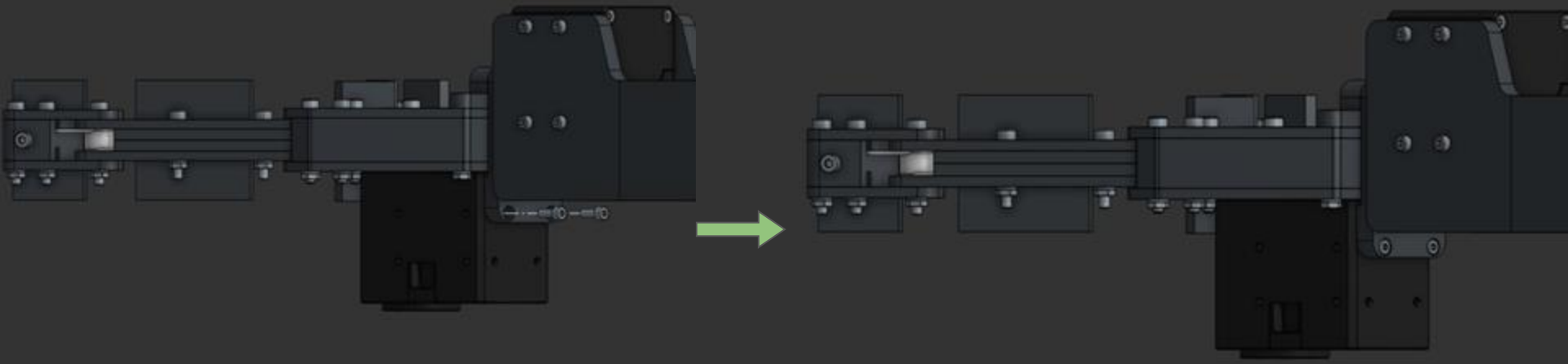
Bolt the Upper Wrist Mount to the Center Wrist Mount



Parts
-Previous step
-2x M2.5x14mm Socket Head Bolt
-2x M2.5 Lock Washer
-2x M2.5 Nut

Step 4

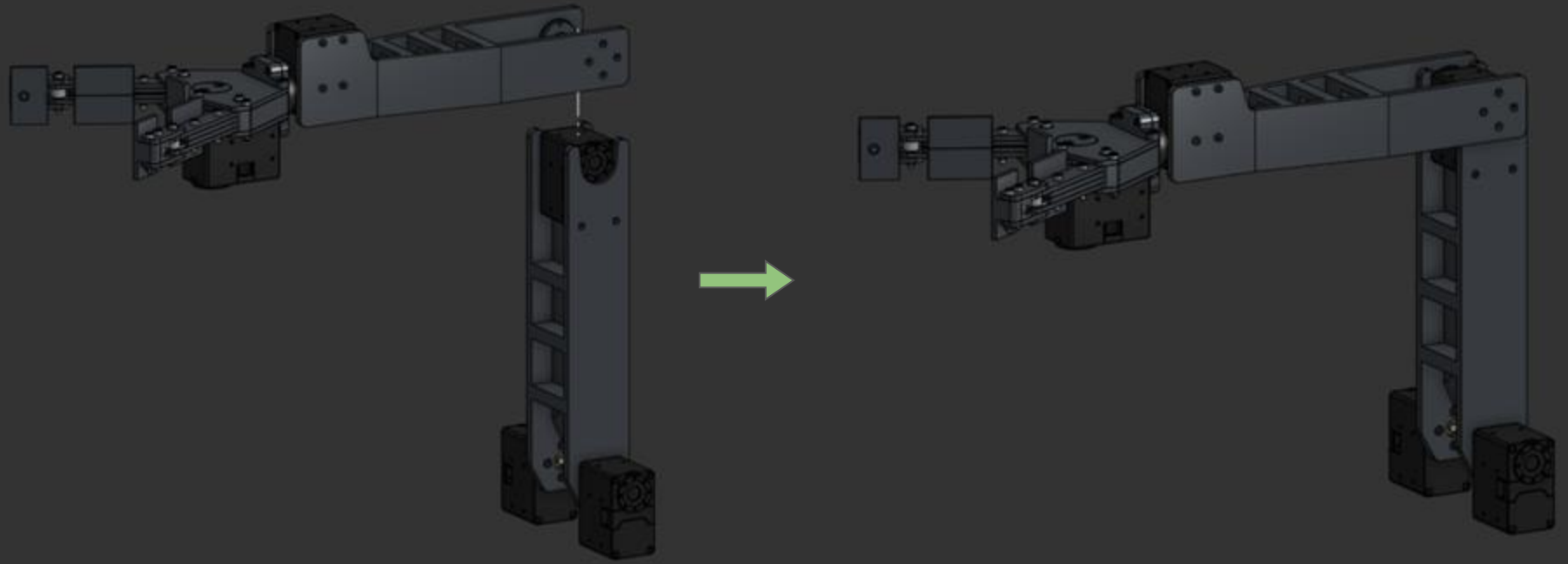
Bolt the bottom of the Center Wrist Mount to the gripper actuator



Parts
-Previous step
-2x M2.5x8mm Socket Head Bolt

Step 5

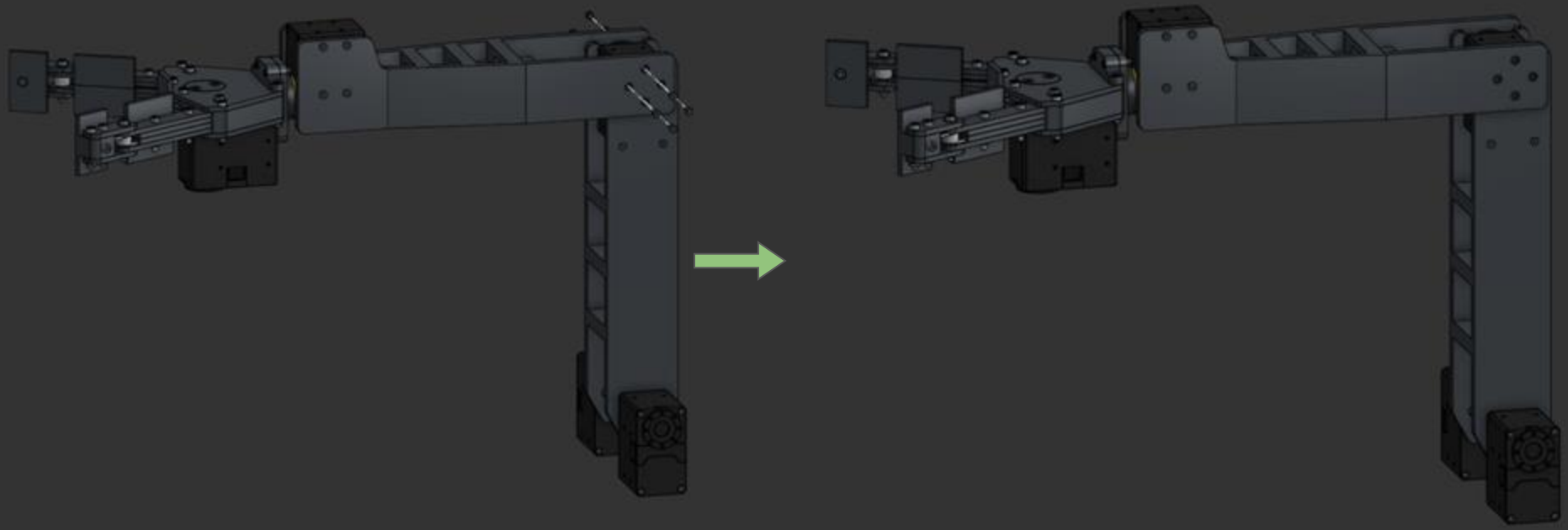
Align the Elbow Assembly with the Shoulder Assembly



Parts
-Previous step
-Shoulder Assembly

Step 6

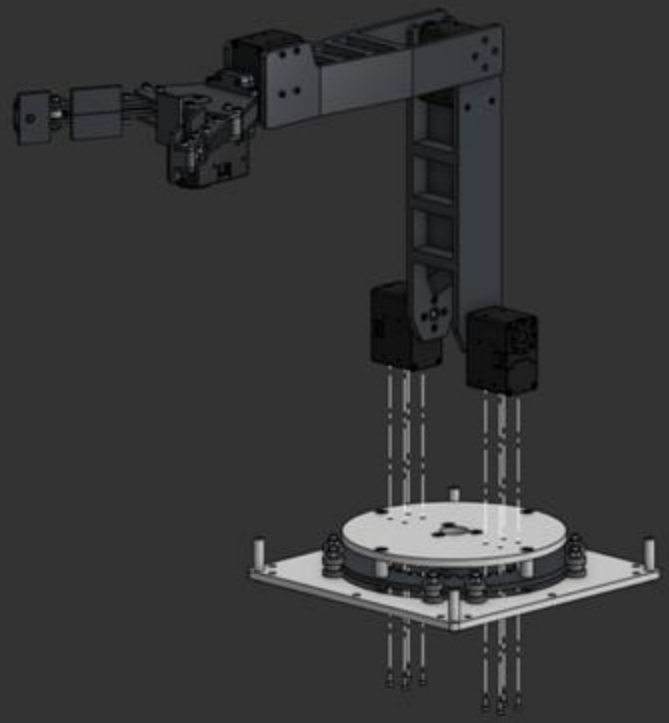
Bolt the Elbow Assembly to the Shoulder Assembly

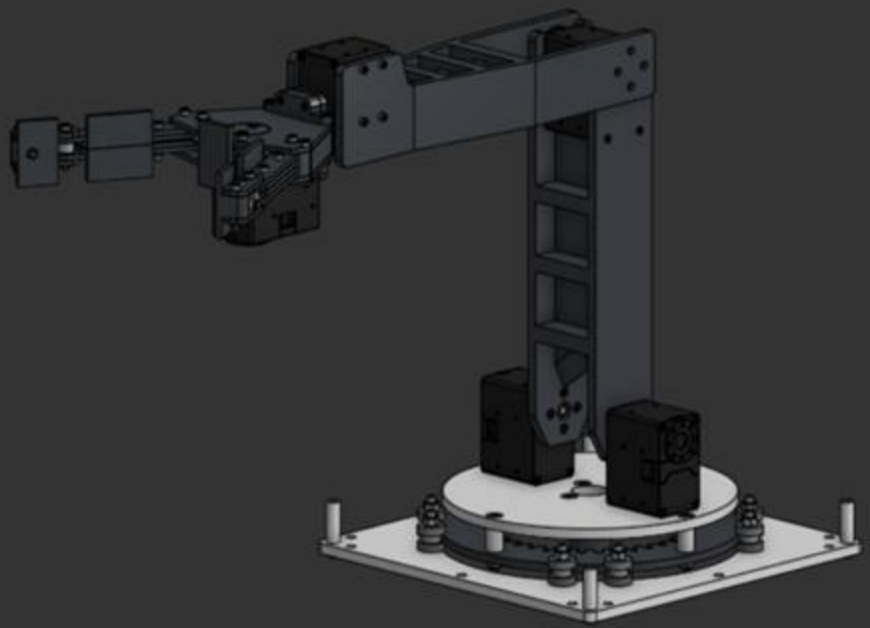


Parts
-Previous step
-8x M2.5x6mm Socket Head Bolt

Step 7

Bolt the Shoulder Assembly to the Arm Base Assembly

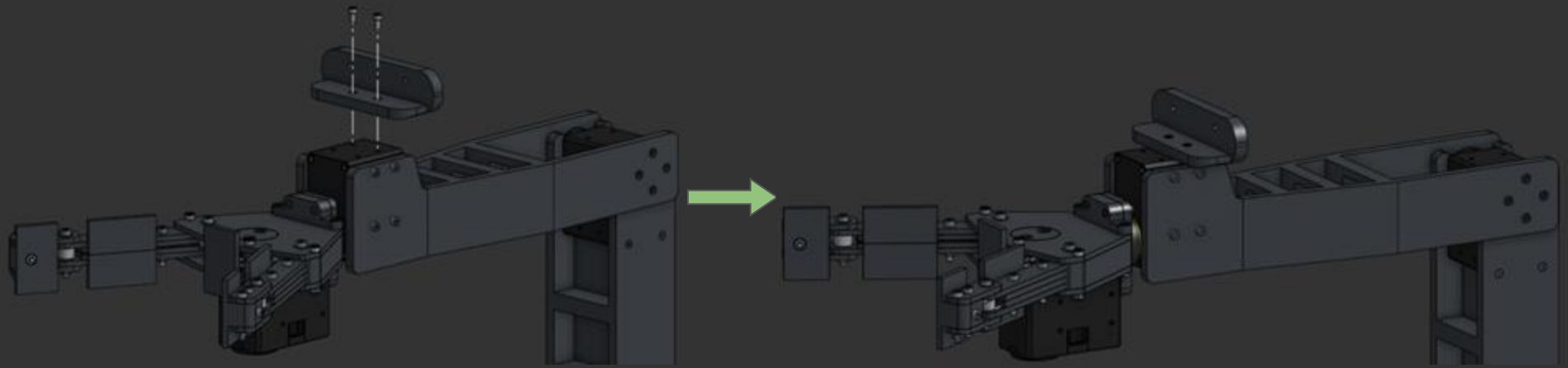




Parts
-Previous step
-Arm Base Assembly
-8x M2.5x8mm Socket Head Bolt

Step 8

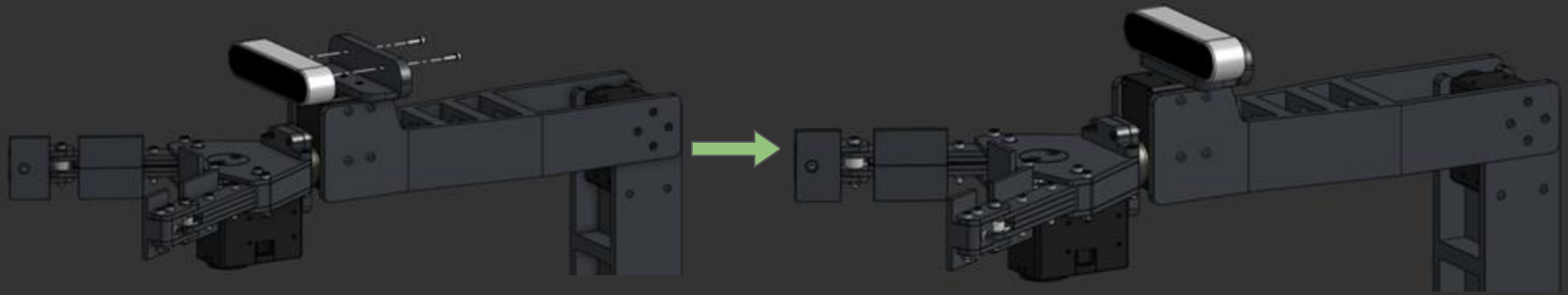
Bolt the Camera Mount to the back two holes of the elbow actuator



Parts
-Previous step
-Camera Mount
-2x M2.5x6mm Socket Head Bolt

Step 9

Bolt the camera to the Camera Wrist Mount



Parts
-Previous step
-Intel Realsense D435i Camera
-2x M3x10mm Socket Head Bolt